UNIT - I

1 Introduction to Soft Computing

1.1 : Introduction to Soft Computing : Evolutionary Computing

Q.1 Define soft computing.

Ans. : • **Definition :** "Soft computing is an emerging approach to computing which parallel the remarkable ability of human mind to reason and learn in a environment of uncertainty and imprecision".

• Soft computing is a collection of methodologies that aim to exploit the tolerance for imprecision and uncertainty to achieve tractability, robustness, and low solution cost

Q.2 Explain unique property of soft computing.

Ans. : Property are :

a. Learning from experimental data

b. Soft computing techniques derive their power of generalization from approximating

c. Generalization is usually done in a high-dimensional space

Q.3 What are the techniques used in soft computing ?

Ans. : • Techniques used in soft computing are neural networks, support vector machines, fuzzy logic and genetic algorithms in evolutionary computation.

Q.4 What are the goal of soft computing ?

Ans. : • It is a new multidisciplinary field, to construct a new generation of Artificial Intelligence, known as computational intelligence.

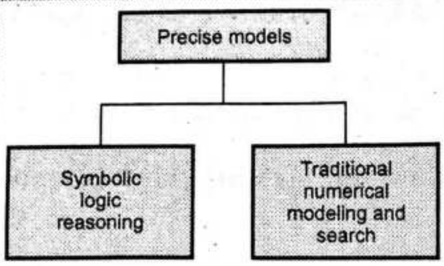
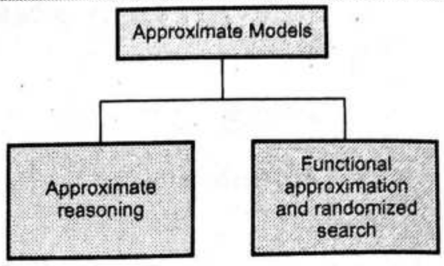
• The main goal is to develop intelligent machines to provide solutions to real world problems, which are not modeled or too difficult to model mathematically.

• Its aim is to develop the tolerance for Approximation, Uncertainty, Imprecision, and Partial Truth in order to achieve close resemblance with human like decision making.

Q.5 Compare hard computing and soft computing.

Ans. :

Sr. No.	Hard computing	Soft computing
1.	It uses binary logic.	It uses fuzzy logic.
2.	It is based on numerical analysis.	It is based on genetic algorithms.
3.	Crisp system is used in had computing.	Neurocomputing is used in soft computing.

4.	It takes help of differential equations.	It takes help of probabilistic reasoning.
5.	Approximation theory.	Evidential reasoning.
6.	Functional analysis.	Machine learning.
7.		

Q.6 List strength and weakness of soft computing.

Ans. : Strength :

1. Low cost and less time consuming solutions are provided by soft computing.
2. Soft computing techniques can tolerate imprecision, uncertainty and partial truth to produce High Machine Intelligent Quotient.
3. It provides good solutions.
4. It uses human concept.

Weakness :

1. It requires extensive computation.
2. Soft computing does not perform much symbolic manipulation.
3. Soft computing is not precisely defined.

Q.7 What is evolutionary computing ? Explain.

Ans. :
- Evolutionary computation is an artificial intelligence subfield and closely linked to computational intelligence, involving lots of combinatorial optimization problems and continuous optimization.
- It is employed in problem-solving systems that use computational models with evolutionary processes as the key design elements. It is an abstraction from the evolutionary concept in biology since it deals with methods and concepts that are continually and selectively evolving and optimizing.
- Evolutionary algorithms are of two types: genetic algorithm and evolutionary strategies.
- Evolutionary algorithms are concerned with investigating computational systems that resemble simplified versions of the processes and mechanisms of evolution toward achieving the effects of these processes and mechanisms, namely the development of adaptive systems

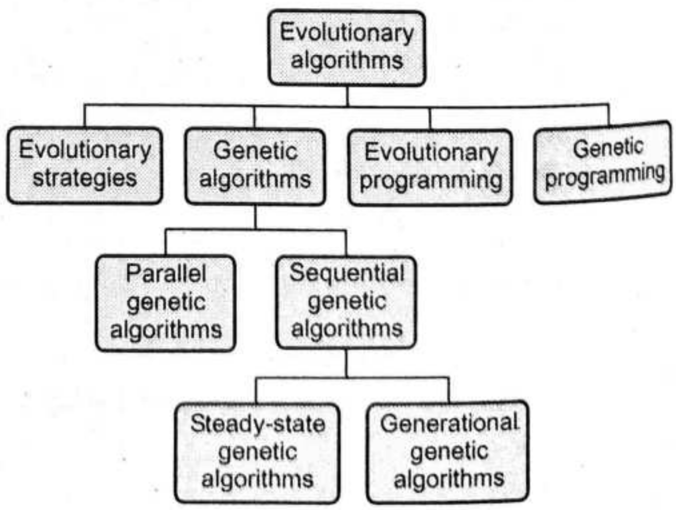


Fig. Q.7.1

1.2 : Soft Computing Methods

Q.8 Define the term neural network.

Ans. : Neural network is a system composed of many simple processing elements operating in parallel whose function is determined by network structure, connection strengths, and the processing performed at computing elements or nodes.

Q.9 List the soft computing methods.

Ans. : Soft Computing Methods are fuzzy set and fuzzy logic, artificial neural network, genetic algorithm, rough set theory etc.

Q.10 Describe fuzzy approach of soft computing.

Ans. : • Fuzzy Logic (FL) is a multivalued logic, that allows intermediate values to be defined between conventional evaluations like true/false, yes/no, high/low, etc.

- Fuzzy Logic is based on fuzzy set theory and provides methods for modeling and reasoning under uncertainty, a characteristic present in many problems, which makes FL a valuable approach.
- It allows data to be represented in intuitive linguistic categories instead of using precise (crisp) numbers which might not be known, necessary or in general may be too restrictive.
- Fuzzy logic offers a practical way for designing nonlinear control systems. It achieves nonlinearly through piece-wise linear approximation. The basic building blocks of a fuzzy logical control system are set of fuzzy if-then (i.e., fuzzy rule based models) that approximate a functional mapping.
- Fuzzy logic provides an inference morphology that enables approximate human reasoning capabilities to be applied to knowledge-based systems. The theory of fuzzy logic provides a mathematical strength to capture the uncertainties associated with human cognitive processes, such as thinking and reasoning.
- Fuzzy systems are suitable for uncertain or approximate reasoning, especially for the system with a mathematical model that is difficult to derive.
- Fuzzy logic allows decision making with estimated values under incomplete or uncertain information.
- Fuzzy logic is viewed as a formal mathematical theory for the representation of uncertainty. Uncertainty is crucial for the management of real systems: if you had to park your car precisely in one place, it would not be possible. Instead, you work within, say, 10 cm tolerances.
- The presence of uncertainty is the price you pay for handling a complex system. Nevertheless, fuzzy logic is a mathematical formalism, and a membership grade is a precise number. What's crucial to realize is that fuzzy logic is a logic of fuzziness, not a logic which is itself fuzzy.

Q.11 List and explain in brief constituents of soft computing.

Ans. : • The principal constituents, i.e., tools, techniques, of Soft Computing (SC) are Fuzzy Logic (FL), Neural Networks (NN),

Support Vector Machines (SVM), Evolutionary Computation (EC), and Machine Learning (ML) and Probabilistic Reasoning (PR).

- Fuzzy theory plays a leading role in soft computing and this stems from the fact that human reasoning is not crisp and admits degrees.

Genetic algorithms

- Genetic algorithms are inspired by Darwin's theory of natural evolution. In the natural world, organisms that are poorly suited for an environment die off, while those well-suited, prosper.
- Genetic algorithms search the space of individuals for good candidates. The chance of an individual's being selected is proportional to the amount by which its fitness is greater or less than its competitors' fitness.

Neural Networks

- Neural networks consists of many number of simple elements (neurons) connected between them in system. Whole system is able to solve of complex tasks and to learn for it like a natural brain.

- For user NN is black box with Input vector (source data) and Output vector (result).

Probabilistic reasoning

- Uncertainty is described by probabilities. Probability may be use for simulation of fuzziness.
- Relations between events are described as conditional probabilities (Bayesian nets) or probabilities of transition probabilities (Markovian process).

Q.12 Difference between digital computer and neural network.

Ans. :

Sr. No.	Digital computer	Neural network
1.	Deductive reasoning : We apply known rules to input data to produce output.	Inductive reasoning : Given input and output data (training examples), we construct the rules.
2.	Computation is centralized, synchronous and serial.	Computation is collective, asynchronous and parallel.
3.	Memory is packetted, literally stored and location addressable.	Memory is distributed, internalized and content addressable.
4.	Not fault tolerant. One transistor goes and it no longer works.	Fault tolerant, redundancy and sharing of responsibilities.
5.	Fast. Measured in millionths of a second.	Slow. Measured in thousandths of a second.
6.	Exact.	Inexact.
7.	Static connectivity.	Dynamic connectivity.
8.	Applicable if well defined rules with precise input data.	Applicable if rules are unknown or complicated or if data is noisy or partial.

Q.13 What is hybrid system ? Explain classification of hybrid system.

Ans. : • In hybrid system, more than one technology is used. Integrated architectures for machine learning have been shown to provide performance improvements over single representation architectures.

- The combination of knowledge based systems, neural networks and evolutionary computation forms the core of an emerging approach to building hybrid intelligent systems capable of reasoning and learning in an uncertain and imprecise environment.
- In recent years multiple module integrated machine learning systems have been developed to overcome the limitations inherent in single component systems. Integrations of neural networks, fuzzy logic and global optimization algorithms have received considerable attention but increasing attention is being paid to integrations with case based reasoning and rule induction.
- Classification of hybrid systems :
  1. Sequential hybrid systems
  2. Auxiliary hybrid systems
  3. Embedded hybrid systems
- Sequential hybrid system uses pipeline concept. The output of one technology is used as input of other technology.
- Auxiliary hybrid system works like subroutine. First technology process or manipulate data required by it and second technology processes the information provided by the first technology and return it to first technology.
- Two technologies is integrated such way that they appear intertwined. Embedded system is hybrid of hardware and software. Neuro-Fuzzy system is an example of embedded hybrid system.

1.3 : Characteristics and Applications of Soft Computing Techniques

Q.14 List and characterize the constituents of soft computing.

Ans. :

1. One of the characteristics of soft computing methods is that they are typically used in problems where mathematical models are not available or are intractable or too cumbersome to be viable.
2. Another characteristic is that uncertainty inherent in many situations under study is taken into account rather than ignored.
3. Soft computing often provides a good solution as opposed to an optimal solution.
4. Soft computing uses human concept like if-then rules, cases and conventional knowledge representations.
5. Biologically inspired computing models (NN).
6. New optimization techniques are applied by soft computing.
7. Model free learning : Models are constructed based on the target system only.
8. New application domains : Mostly computation intensive like adaptive signal processing, adaptive control, nonlinear system identification etc.
9. Fault tolerance : Deletion of a neuron or a rule does not destroy the system. The system performs with lesser quality.
10. Goal driven characteristics : Only the goal is important and not the path.

Q.15 What type of problem is solved by using soft computing ?

Ans. : • Soft computing techniques have become one of promising tools that can provide practice and reasonable solution

1. Feature Selection : • In machine learning and statistics, feature selection, also known as variable selection, attribute selection or variable subset selection, is the process of selecting a subset of relevant features for use in model construction.

- Feature selection techniques are a subset of the more general field of feature extraction. Feature extraction creates new features from functions of the original features, whereas feature selection returns a subset of the features.

2. Image Processing : • In imaging science, image processing is any form of signal processing for which the input is an image, such as a photograph or video frame; the output of image processing may be either an image or a set of characteristics or parameters related to the image.

- Most image-processing techniques involve treating the image as a two-dimensional signal and applying standard signal processing techniques to it.

3. Medical Diagnosis : • Medical diagnosis refers both to the process of attempting to determine or identify a possible disease and to the opinion reached by this process. From the point of view of statistics the diagnostic procedure involves classification tests.

4. Pattern Recognition : • Pattern recognition generally aim to provide a reasonable answer for all possible inputs and to perform "most likely" matching of the inputs, taking into account their statistical variation.

- Pattern recognition is studied in many fields, including psychology, psychiatry, and ethology, cognitive science, and traffic flow and computer science.

5. Smart Instrumentation : • As intelligent devices become ubiquitous the challenge is to connect sensors and actuators through smart systems. A major issue is the growing number of communication protocols, with no single standard.

- The challenge is to intelligently connect smart instrumentation so that devices can communicate across multiple protocols. At the same time, increases in the volume and importance of data means that privacy, security and robustness of systems is paramount.

Fill in the Blanks for Mid Term Exam

Q.1 ________ is a collection of methodologies that aim to exploit the tolerance for imprecision and uncertainty to achieve tractability, robustness and low solution cost.

Q.2 Soft computing is a concept that was introduced by ________, the discoverer of fuzzy logic.

Q.3 Hard computing is also called as ________ computing.

Q.4 Soft computing uses ________ logic.

Q.5 Hard computing uses ________ logic.

Mutliple Choice Questions for Mid Term Exam

Q.1 Application of Neural network includes _____.

a pattern recognition
b classification
c clustering
d all of these

Q.2 Neural networks are complex ________ with many parameters

a linear functions
b nonlinear functions
c discrete functions
d exponential functions

Q.3 Types of soft computing techniques are ________

a fuzzy logic
b neural network
c genetic algorithm
d all of these

Q.4 Soft computing is a concept that was introduced by ________, the discoverer of fuzzy logic.

a Lotfi A. Zadeh
b Rechenberg
c Newton
d Charles Darwin

Q.5 Soft computing is based on ________

a fuzzy logic
b neural science
c crisp software
d binary logic

Q.6 Hard computing uses ________ logic.

a soft
b binary
c fuzzy
d all of these

Q.7 Neural network needs self organizing capability for ________ learning.

a reinforcement
b rote
c supervised
d unsupervised

Answer Keys for Fill in the Blanks :

Q.1	Soft computing	Q.2	Zadeh
Q.3	conventional	Q.4	fuzzy
Q.5	binary		

Answer Keys for Multiple Choice Questions :

Q.1	d	Q.2	a
Q.3	d	Q.4	a
Q.5	a	Q.6	b
Q.7	d		

END...

2 Fuzzy Systems

2.1 : Fuzzy Set

Q.1 What is fuzzy set ?

Ans. : • A fuzzy set is a collection of objects with graded membership. Two examples of "Sets" :

1. All employees of XYZ who are over 1.8 meter in height.
2. All employees of XYZ who are tall.

- The first example is a **classical set,** we have a universe (all XYZ employees) and a membership rule that divides the universe into members (those over 1.8 m) and nonmembers.
- The second example is a **fuzzy set,** some employees are definitely in the set and some are definitely not in the set, but some are "borderline". This distinction between the "ins", the "outs", and the "borderlines" is made more exact by the membership function, $\mu_A(x)$.

Q.2 How to represent a fuzzy set in a computer ?

Ans. : The membership function must be determined first. A number of methods learned from knowledge acquisition can be applied here. For example, one of the most practical approaches for forming fuzzy sets relies on the knowledge of a single expert.

Q.3 Give the properties of fuzzy sets and also explain the operations involved in it.

Ans. : Properties of fuzzy sets are as follows : Let A, B and C be fuzzy sets in a universe of discourse U, then we have,

Idempotent laws	$A \cup A = A$, $A \cap A$
Commutative laws	$A \cup B = B \cup A$, $A \cap B = B \cap A$
Associative laws	$(A \cup B) \cup C = A \cup (B \cup C)$, $(A \cap B) \cap C = A \cap (B \cap C)$
Absorption laws	$A \cup (A \cap B) = A$, $A \cap (A \cup B) = A$
Distributive laws	$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$, $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
Involution law	$\overline{\overline{A}} = A$
De Morgan's laws	$\overline{A \cup B} = \overline{A} \cap \overline{B}$, $\overline{A \cap B} = \overline{A} \cup \overline{B}$
Complement laws	$A \cup \overline{A} \neq U$, $A \cap \overline{A} \neq \varnothing$

Operations on Fuzzy Sets

- Union, intersection, and complement are the most basic operations on classical sets. On the basis of these three operations, a number of identities can be established. Corresponding to the ordinary set operations of union, intersection, and complement, fuzzy sets have similar operations.

1. **Containment or subset :** Fuzzy set A is **contained** in fuzzy set B (or, equivalently, A is a **subset** of B, or A is smaller than or equal to B, $A \subseteq B$) if and only if $\mu_A(x) \le \mu_B(x)$ for all x.

$$A \subseteq B \Leftrightarrow \mu_A \le \mu_B$$

A is contained in B

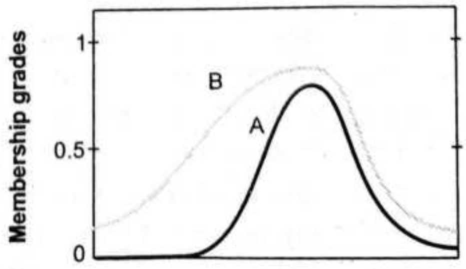


Fig. Q.3.1 Concept of $A \subseteq B$

2. **Union (disjunction) :** The **union** of two fuzzy sets A and B is a fuzzy set C, written as $C = A \cup B$ or C = A OR B, whose MF is related to those of A and B by $\mu_C(x) = \max(\mu_A(x), \mu_B(x))$.

$$C = A \cup B \Leftrightarrow \mu_c(x) = \max\ (\mu_A(x), \mu_B(x)) = \mu_A(x) \vee \mu_B(x)$$

3. **Intersection (conjunction) :** The **intersection** of two fuzzy sets A and B is a fuzzy set C, written as $C = A \cap B$ or C = A AND B, whose MF is related to those of A and B by $\mu_C(x) = \min(\mu_A(x), \mu_B(x))$.

$$C = A \cap B \Leftrightarrow \mu_C(x) = \min(\mu_A(x), \mu_B(x)) = \mu_A(x) \wedge \mu_B(x)$$

4. **Complement (negation) :** The complement of fuzzy set A, denoted by $\overline{A}$ or NOT A, is defined as $\mu_A(x) = 1 - \mu_A(x)$.

5. **Cartesian product and co-product :** Let A and B be the fuzzy sets in X and Y respectively. The cartesion product of A and B denoted by $A \times B$, is a fuzzy set in the product space $X \times Y$ with the membership function

$$\mu_{A \times B}(x, y) = \min(\mu_A(x), \mu_B(x)).$$

Cartesian co-product A + B is a fuzzy set with the membership function

$$\mu_{A+B}(x, y) = \max(\mu_A(x), \mu_B(x)).$$

Fig. Q.3.2 shows the basic three operation on two fuzzy set A and B.

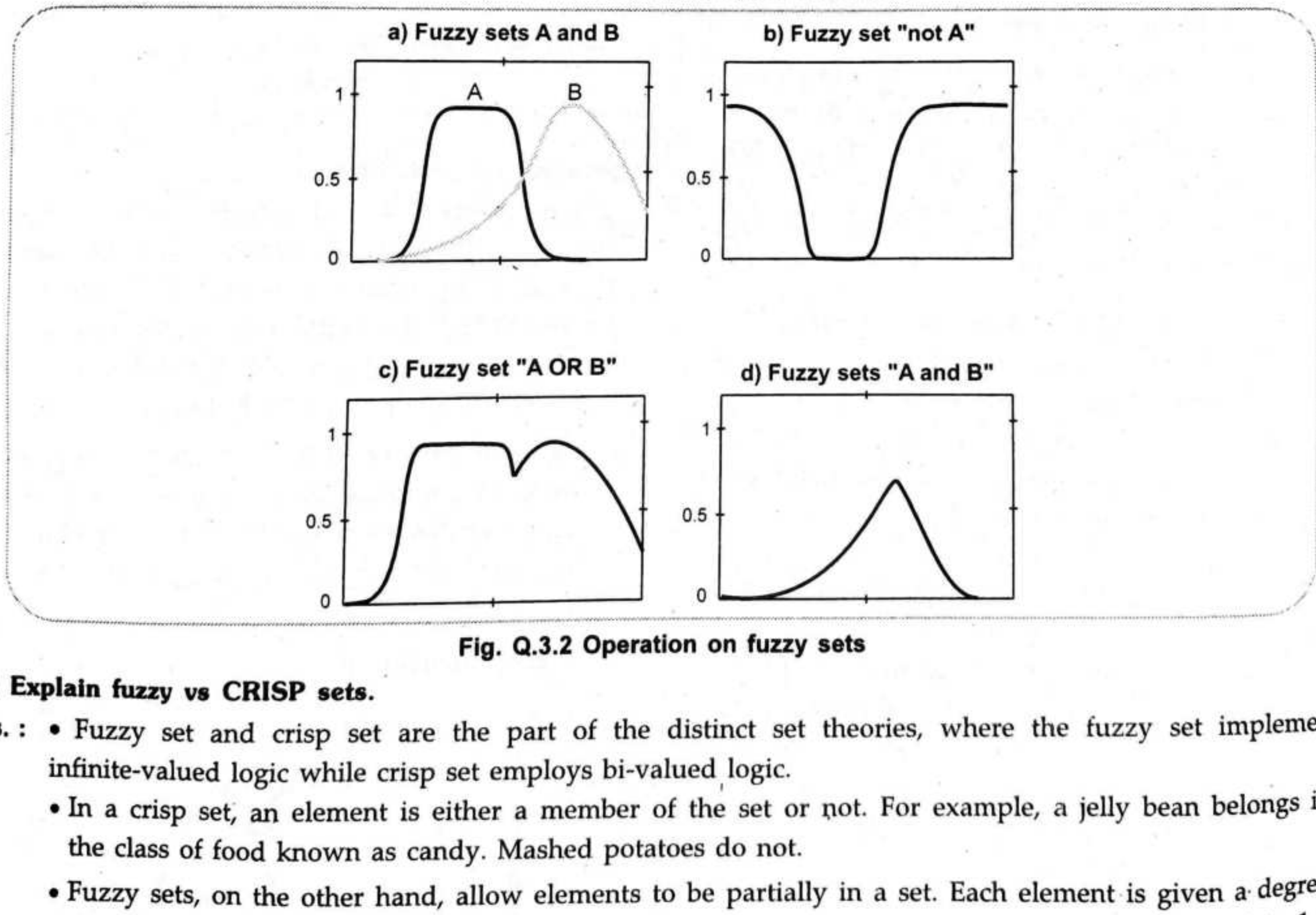


Fig. Q.3.2 Operation on fuzzy sets

Q.4 Explain fuzzy vs CRISP sets.

Ans. : • Fuzzy set and crisp set are the part of the distinct set theories, where the fuzzy set implements infinite-valued logic while crisp set employs bi-valued logic.

- In a crisp set, an element is either a member of the set or not. For example, a jelly bean belongs in the class of food known as candy. Mashed potatoes do not.
- Fuzzy sets, on the other hand, allow elements to be partially in a set. Each element is given a degree of membership in a set. This membership value can range from 0 (not an element of the set) to 1 (a member of the set). It is clear that if one only allowed the extreme membership values of 0 and 1, that this would actually be equivalent to crisp sets.

Parameters	Crisp set	Fuzzy set
Basic	Defined by precise and certain characteristics	Prescribed by vague or ambiguous properties
Property	Element is either the member of a set or not	Elements are allowed to be partially included in the set
Application	Digital design	Used in fuzzy controllers
Logic	Bi-valued	Infinite-valued

Q.5 Define classical set.

Ans. : A classical set is either an element belongs to the set or it does not. For example, for the set of integers, either an integer is even or it is not (it is odd). However, either you are in the Indian or you are not.

Q.6 What is membership function ?

Ans. : Fuzzy sets are characterized by **membership functions.** A fuzzy set has a graphical description that expresses how the transition from one to another takes place. This graphical description is called a **membership function.**

Q.7 What is alpha cut ?

Ans. : α-cuts reduce a fuzzy set into an extracted crisp set. The value α represents a membership degree, i.e., $\alpha \in [0, 1]$. The α-cut of a fuzzy set A is the crisp set $(A - \alpha)$, i.e., the set of all elements whose membership degrees in A are $\geq \alpha$.

Q.8 Define membership function.

Ans. : Membership function is a function from a universal set U to the interval [0, 1]. A fuzzy set A is defined by its membership function ϕ_A over U.

Q.9 What is cardinality of fuzzy set ?

Ans. : The cardinality of fuzzy sets is a measure of the number of the elements belonging to the set.

Q.10 What is degree of membership ?

Ans. : For any element x of universe X, membership function $\mu A(x)$ equals the degree to which x is an element of set A. This degree, a value between 0 and 1, represents the degree of membership, also called membership value, of element x in set A.

Q.11 Explain following fuzzy set operations with example :

a. Empty fuzzy set b. Product fuzzy set c. Normal fuzzy set

Ans. : • Let X be a nonempty set. A fuzzy set A in X is characterized by its membership function

$$\mu_A : X \rightarrow [0, 1]$$

and $\mu_A(x)$ is interpreted as the degree of membership of element x in fuzzy set A for each

$x \in X$.

- It is clear that A is completely determined by the set of tuples

$$\wedge = \{(u, \mu_A(u)) | u \in X\}$$

Frequently we will write A(x) instead of $\mu_A(x)$. The family of all fuzzy sets in X is denoted by F (X).

If $X = \{x_1, \ldots, x_n\}$ is a finite set and A is a fuzzy set in X then we often use the notation

$$A = \mu_1/x_1 + \ldots + \mu_n/x_n$$

where the term μ_i/x_i, $i = 1, \ldots, n$ signifies that μ_i is the grade of membership of x_i in A and the plus sign represents the union.

i) **Normal fuzzy set :** A fuzzy subset A of a classical set X is called normal if there exists an $x \in X$ such that $A(x) = 1$. otherwise A is subnormal.

ii) **Product fuzzy Set :** Let A and B be the fuzzy sets in X and Y respectively. The cartesion product of A and B denoted by A x B, is a fuzzy set in the product space X x Y with the membership function

$$\mu_{A \times B}(x, y) = \min(\mu_A(x), \mu_B(x)).$$

Cartesian co-product $A + B$ is a fuzzy set with the membership function

$$\mu_{A+B}(x, y) = \max(\mu_A(x), \mu_B(x)).$$

iii) **Empty fuzzy set :** The empty fuzzy subset of X is defined as the fuzzy subset ϕ of X such that

$$\phi(x) = 0 \text{ for each } x \in X$$

It is easy to see that $\phi \subset A$ holds for any fuzzy subset A of X.

Q.12 What is the alpha-cut method for discrete fuzzy sets ?

Ans. : An α cut of a fuzzy set A is a crisp set ${}^{\alpha}A$ that contains all the elements in X that have membership value in A strictly greater than or equal to α.

$${}^{\alpha}A - \{x \mid A(x) \geq \alpha\}$$

- A strong α-cut of a fuzzy set A is crisp set ${}^{\alpha}A$ that contains all the elements in X that have membership value in A strictly greater than α.

$${}^{\alpha+}A - \{x \mid A(x) > \alpha\}$$

- α-cuts reduce a fuzzy set into an extracted crisp set. The value α represents a membership degree, i.e. $\alpha \in [0,1]$. The α-cut of a fuzzy set A is the crisp set $(A-\alpha)$ i.e., the set of all elements whose membership degress in A are $\geq \alpha$.

Q.13 What is membership functions ? How fuzzy membership function is represented ?

Ans. : • Fuzzy set is any set that allows its members to have different degree of membership, called membership function in the interval [0, 1].

- Fuzzy sets are characterized by Membership Functions. A fuzzy set has a graphical description that expresses how the transition from one to another takes place. This graphical description is called a membership function.
- The **membership function** assigns to each element x in a fuzzy set a number, A(x), in the closed unit interval [0, 1]. The number, A(x), represents the degree of membership of x in A. Hence, membership functions are functions of the form : $A : X \rightarrow [0, 1]$.
- In the case of crisp sets, the members of a set are either out of the set, with membership degree of zero, or in the set, with the value one being the degree of membership.
- Therefore, Crisp Sets $\subseteq$ Fuzzy Sets or in other words, **crisp sets are special cases of fuzzy sets.**
- An element can be in the set with a degree of membership and out of the set with a degree of membership.
- Hence, crisp sets are a special case, or a subset, of fuzzy sets, where elements are allowed a membership degree of 100 % or 0 % but nothing in between.
- There are four ways of representing fuzzy membership functions, namely,
  a. Graphical representation,
  b. Tabular and list representation,
  c. Geometric representation,
  d. Analytic representation
- Tabular and list representations are used for finite sets.
- The third method of representation is the geometric representation and is also used for representing finite sets.
- Analytical representation is another alternative to graphical representation in representing infinite sets, e.g. a set of real numbers.
- The specification of membership functions is subjective, which means that the membership functions specified for the same concept by different persons may vary considerably.

Q.14 List and explain methods employed for membership value assignment.

Ans. : Membership value assignments methods :

1. Intuition
2. Inference
3. Rank ordering
4. Neural networks

- **Intuition :** Intuition involves contextual and semantic knowledge about an issue; it can also involve linguistic truth values about this knowledge.
- **Inference :** In the inference method we use knowledge to perform deductive reasoning. Let A, B, and C be the inner angles of a triangle, in the order $A \geq B \geq C \geq 0$, and let U be the universe of triangles, i.e

$$U = \{(A, B, C) \mid A \geq B \geq C \geq 0; A + B + C\}$$
$$= 180^{\circ}$$

- **Rank ordering :** Assessing preferences by a single individual, a committee, a poll, and other opinion methods can be used to assign membership values to a fuzzy variable. Preference is determined by pair-wise comparisons, and these determine the ordering of the membership.

- **Neural network :** It is a technique that seeks to build an intelligent program using models that simulate the working network of the neurons in the human brain. In a global sense, a neuron receives a set of input pulses and sends out another pulse that is a function of the input pulses. Neural systems solve problems by adapting to the nature of the data(signals) they receive.
- Once the neural network is trained and verified to be performing satisfactorily, it can be used to find the membership of any other data points in the two fuzzy classes.
- A complete mapping of the membership of different data points in the different fuzzy classes can be derived to determine the overlap of the different classes.

2.2 : Fuzzy Relations

Q.15 Define and explain classical relations and fuzzy relations.

Ans • Relations are the way in which two or more things are connected. In natural language, relations are kinds of links existing between objects such as "mother of", "neighbor of", "part of", etc.

- Mathematically, a relation implies the presence of an association between elements of different sets (at least two fuzzy sets). Binary relation is a relation between two sets. Tertiary relation is a relation between three sets and n-ary relation is a relation between n elements.
- Classical relations are structure that represents the presence or absence of correlation or interaction among various sets. Two degrees of relationship in a crisp relations : *Completely related* and *Not related.*
- A binary relation is a relation between two sets X and Y denoted by R(X,Y). A binary fuzzy relation can be expressed as
  1. Linguistically, i.e. through statement
  2. By listing the set of all ordered pairs
  3. Directed graph
  4. Matrix relation R
  5. Table
- **A fuzzy relation is** based on the philosophy that everything is related to some extend or unrelated. Unlike crisp relations, the strength of the relation between ordered pair of two universes is not measured with the characteristic function, but rather with a membership function expression expressing various degree of strength of the relation on the unit interval [0,1].
- Crisp relations and fuzzy relations can be defined in terms of subsets.
- A binary fuzzy relation is a fuzzy relation between two sets X and Y denoted by R(X,Y). A binary fuzzy relation can be expressed as
  a. By listing the set of all ordered pairs
  b. Directed graph
  c. Matrix relation R
  d. Table (Tabular form)
- Let X and Y be two universes of discourse, then

$$R = \{ ((x, y), \mu_R (x, y)) \mid (x, y) \in X \times Y\}$$

is a **binary fuzzy relation in $X \times Y$.**

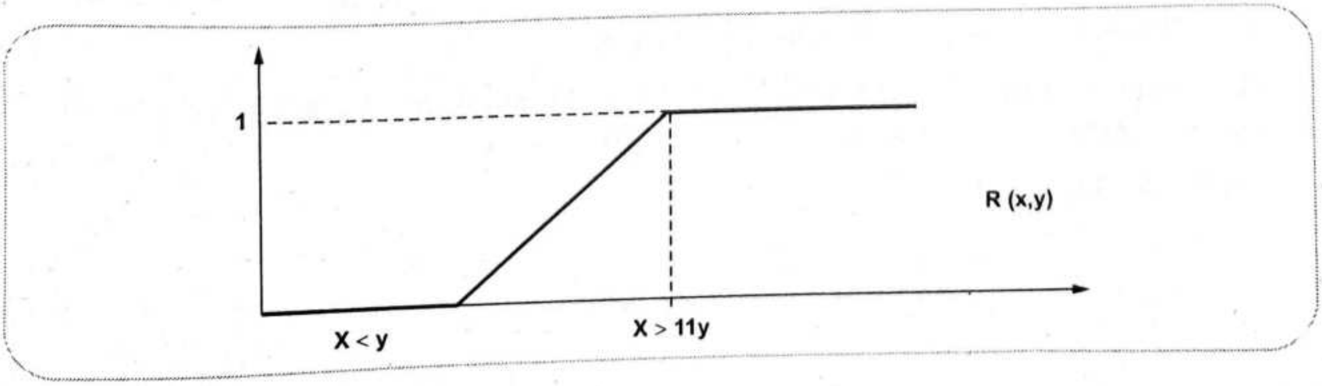


Fig. Q.15.1

- Fuzzy relations represents the strength of association between elements of the two sets.

Example : R = "x is considerably larger than y"

R (X, Y) = Relation between sets X and Y

R (x, y) = Membership function for the relation R (X, Y)

$$R(X, Y) = \{R(x, y) / (x, y) \setminus (x, y) \in (X \times Y)\}$$

$$R(x, y) = \begin{cases} 0 & \text{for } x \leq y \\ (x-y)/(10-y) & \text{for } y < x \leq 11\,y \\ 1 & \text{for } x > 11\,y \end{cases}$$

- Let X = Y = R+ and R = "y is much greater than x" . The member function of the fuzzy relation R can be subjectively defined as

$$\mu_R(x, y) = \begin{cases} \dfrac{y-x}{x+y+2} & \text{if } y\,.\,x \\ 0 & \text{if } y \leq x \end{cases}$$

Q.16 Differentiate between crisp relations and fuzzy relations.

Ans. : • Crisp Relation R from a set A to a set B assigns to each ordered pair exactly one of the following statements "a is related to b" or "a is not related to b".

- Fuzzy relations map elements of one universe (U) to those of another universe (V) through the Cartesian product of the two universes. However, the "strength" of the relation between ordered pairs of the two universes is measured with a membership function expressing various "degrees" of strength of the relation on the unit interval [0,1].

Q.17 What are fuzzy relations ? Explain following operation on fuzzy relations.
i) Intersection ii) Containment

Ans. : • Union, intersection, and complement are the most basic operations on classical sets. On the basis of these three operations, a number of identities can be established.

- Corresponding to the ordinary set operations of union, intersection, and complement, fuzzy sets have similar operations.

Intersection (conjunction) : The intersection of two fuzzy sets A and B is a fuzzy set C, written as $C = A \cap B$ or C = A And B, whose MF is related to those of A and B by $\mu_C(x) = \min(\mu_A(x), \mu_B(x))$

$$C = A \cap B \Leftrightarrow \mu_C(x) = \min(\mu_A(x), \mu_B(x)) = \mu_A(x) \wedge \mu_B(x)$$

ii) Containment : Fuzzy set A is **contained** in fuzzy set B (equivalently, A is a **subset** of B, or A is smaller than or equal to B, $A \subseteq B$) if and only if $\mu_A(x) \leq \mu_B(x)$ for all x.

$$A \subseteq B \Leftrightarrow \mu_A \leq \mu_B$$

- A is contained in B

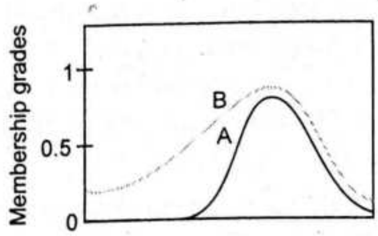


Fig. Q.17.1 Concept of A ⊆ B

Q.18 What is T-norm, T-conorm? Explain order of T-norms.

Ans. :

1. T-norm

- A t-norm (triangular norm) is a kind of binary operation used in the framework of probabilistic metric spaces and in multi-valued logic, specifically in fuzzy logic. A t-norm generalizes intersection in a lattice and conjunction in logic.
- Fuzzy intersection of two fuzzy sets can be specified in a more general way by a binary operation on the unit interval, i.e., a function of the form :

$$T : [0, 1] \times [0, 1] \rightarrow [0, 1]$$

- In order for a functionTto qualify as a fuzzy intersection, it must have appropriate properties. Functions known ast-norms (triangular norms) posses the properties re-quired for the intersection. Similarly, functions calledt-conorms can be used for the fuzzy union.
- A t-norm T is a binary operation on the unit interval that satisfies at least the following axioms for all a; b; c ∈ [0, 1].
- Basic t-norms have following order:

$$t_{min}\,(a, b) \le \max(0, a+b-1) \le ab \le \min(a, b)$$

(T1) $T(x, y) = T(y, x)$,

i.e., the t-norm is commutative,

(T2) $T(T(x, y), z) = T(x, T(y, z))$,

i.e., the t-norm is associative

(T3) $x \le y \Rightarrow T(x, z) \le T(y, z)$,

i.e., the t-norm is monotone,

(T4) $T(x, 1) = x$,

i.e., a neutral element exists, which is 1.

- **Some frequently used t-norms are :**
  1. Standard (Zadeh) intersection : **T(a; b) = min (a; b)**
  2. Algebraic product (probabilistic intersection) : **T(a; b) = ab**
  3. Likewise (bold) intersection : **T(a; b) = max (0; a + b − 1)**
- Although continuous t-norms are most commonly used in fuzzy logics and have an elegant representation

2. T-conorm

- A t-conorm S is a binary operation on the unit interval that satisfies at least the following axioms for all a; b; c ∈ [0; 1]
- Some frequently used t-conorms are :
  1. Standard (Zadeh) union: **S(a; b) = max(a; b);**
  2. Algebraic sum (probabilistic union) : **S(a; b) = a + b − ab;**
  3. Like wise (bold) union: **S(a; b) = min(1; a + b)**
- The minimum is the biggest t-norm and maximum is the smallest t-conorm.

2.3 : Fuzzy Logic

Q.19 What is fuzziness ?

Ans. : • For example, if pressure takes values between 0 and 50, one might label the range 20 to 30 as medium pressure.

- Medium is known as a linguistic variable. Therefore, with Boolean logic 15.0 (or even 19.99) is not a member of the medium pressure range. As soon as the pressure equals 20, then it becomes a member.
- Fuzziness is not vague and multi-valued logic.
- Fuzzy logic may be considered as an extension of multi-valued logic but they are somewhat different. Multi-valued logic is still based on exact reasoning whereas fuzzy logic is approximate reasoning.
- Fuzzy logic provides an inference morphology that enables approximate human reasoning capabilities to be applied to knowledge-based systems. The

theory of fuzzy logic provides a mathematical strength to capture the uncertainties associated with human cognitive processes, such as thinking and reasoning.

- The conventional approaches to knowledge representation lack the means for representation the meaning of fuzzy concepts.
- The development of fuzzy logic was motivated in large measure by the need for a conceptual frame work which can address the issue of uncertainty and lexical imprecision.

Q.20 What is meant by fuzzy logic system ? Illustrate it with proper examples.

Ans. : • Fuzzy set theory was developed by Lotfi A. Zadeh, professor for computer science at the University of California in Berkeley, to provide a mathematical tool for dealing with the concepts used in natural language.

- Fuzzy logic is not logic that is fuzzy, but logic that is used to describe fuzziness. Fuzzy logic is the theory of fuzzy sets, sets that calibrate vagueness.
- Fuzzy Logic is basically a multi-valued logic that allows intermediate values to be defined between conventional evaluations. Fuzzy logic is a form of many-valued logic in which the truth values of variables may be any real number between 0 and 1.
- Fuzzy logic is an extension of Boolean logic used to describe environments where there is no absolute truth and there is uncertainty.
- Fuzzy logic was specifically designed to mathematically represent uncertainty and vagueness and to provide formalized tools for dealing with the imprecision intrinsic to many problems.
- Fuzzy logic provides an inference morphology that enables approximate human reasoning capabilities to be applied to knowledge-based systems. The theory of fuzzy logic provides a mathematical strength to capture the uncertainties associated with human cognitive processes, such as thinking and reasoning.
- Fuzzy systems are **knowledge-based or rule-based systems.** The heart of a fuzzy system is a knowledge base consisting of the so-called fuzzy IF-THEN rules.
- FL is a problem-solving control system methodology that lends itself to implementation in systems ranging from simple, small, embedded micro-controllers to large, networked, multi-channel PC or workstation-based data acquisition and control systems.
- The conventional approaches to knowledge representation lack the means for representing the meaning of fuzzy concepts. As a consequence, the approaches based on first order logic and classical probability theory do not provide an appropriate conceptual framework for dealing with the representation of common sense knowledge, since such knowledge is by its nature both lexically imprecise and non-categorical.

What is Fuzziness ?

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- Fuzzy logic provides an inference morphology that enables approximate human reasoning capabilities to be applied to knowledge-based systems. The theory of fuzzy logic provides a mathematical strength to capture the uncertainties associated with human cognitive processes, such as thinking and reasoning.
- The conventional approaches to knowledge representation lack the means for representation the meaning of fuzzy concepts.
- The development of fuzzy logic was motivated in large measure by the need for a conceptual frame work which can address the issue of uncertainty and lexical imprecision.

Q.21 List out the characteristics features of fuzzy logic.

Ans. : • Essential **characteristics of fuzzy logic :**

1. In fuzzy logic, exact reasoning is viewed as a limiting case of approximate reasoning.
2. Everything is a matter of degree in fuzzy logic.
3. Knowledge is interpreted a collection of elastic or, equivalently, fuzzy constraint on a collection of variables.
4. Inference is viewed as a process of propagation of elastic constraints.
5. Any logical system can be fuzzified.

• There are two main characteristics of fuzzy systems that give them better performance for specific applications.

1. Fuzzy systems are suitable for uncertain or approximate reasoning, especially for the system with a mathematical model that is difficult to derive.
2. Fuzzy logic allows decision making with estimated values under incomplete or uncertain information.

• Fuzzy systems are suitable for uncertain or approximate reasoning, especially for the system with a mathematical model that is difficult to derive. Fuzzy logic allows decision making with estimated values under incomplete or uncertain information.

• Fuzzy logic is based on the idea that all things admit of degrees. For example, temperature, height, speed, distance and beauty all come on a sliding scale. The motor is running really hot. Ram is a very tall guy.

Q.22 List the merits and demerits of fuzzy logic.

Ans. : Merits of fuzzy logic

1. Allows the use of vague linguistic terms in the rules.
2. Modularity : Rules can be added and removed as needed.
3. Parallel execution of rules : Output calculated once at end of cycle.
4. Fuzzy logic inherently accounts for noise in the data because it extracts trends, not precise values.
5. Algorithms in fuzzy logic are cast in the same language used in day-to-day conversation which makes fuzzy logic predictions easily interpretable.
6. Fuzzy logic algorithms are computationally efficient and can be scaled to include an unlimited number of components.

Demerits of fuzzy logic

1. Difficult to estimate membership function.
2. There are many ways of interpreting fuzzy rules.
3. Computational cost : More computations involved.
4. Rules need to be expressive and needs to be accurate.

Q.23 "Behavior of fuzzy logic is deterministic"? Justify.

Ans. : • Fuzzy logic can be viewed as a super set of Boolean logic, as a multi-valued logic. It adds degrees between the absolute truth and absolute false to cover partial truth in between.

• In simple terms, fuzzy logic involves classifying objects and functions into fuzzy sets which could be given linguistic phrases.

• It is a form of reasoning that is neither exact, nor absolutely inexact. For example, too hot, little slow, phrases which do not give the idea of absolute, but a fuzzy estimate.

• While 33 degrees might be warm enough for a person from the equator, someone from the arctic might find the heat unbearable, or too hot.

• It is not possible to classify them into strict sets with defined boundaries which leads to the idea of fuzziness.

• It has basically evolved from predicate logic, though many forms called t-norm fuzzy logics do exist in propositional logic too. It generally has an object and a predicate.

• For example : In the sentence, "Plato is a man", 'Plato' is the object, and "is a man" is the predicate. But an important point about fuzzy logic is that it is deterministic and time-variant.

- A few salient points on fuzzy logic are :
  1. Exact reasoning or precise values is the extreme or limiting case of approximate reasoning.
  2. Any system which works on logic can be fuzzified and everything would be a matter of degrees.
  3. Knowledge is a collection of fuzzy constraints on a group of variables.

Q.24 Consider two fuzzy sets A and B

$$A = \left\{\frac{1}{2}+\frac{0.5}{3}+\frac{0.3}{4}+\frac{0.2}{5}\right\}$$

$$B = \left\{\frac{0.5}{2}+\frac{0.7}{3}+\frac{0.2}{4}+\frac{0.4}{5}\right\}$$

Perform the following operating on fuzzy sets

i) $A \cup B$ ii) $A \cap B$ iii) Compement of fuzzy set A iv) Difference $\left(\frac{A}{B}\right)$

v) Algebraic sum of given fuzzy sets. vi) Bounded sum of the given fuzzy set.
vii) Algebraic product of the given fuzzy sets. viii) $\overline{A \cup B}$ ix) $A \cup \overline{B}$

Ans. : i) $A \cup B$

$$A \cup B = \frac{1}{2}+\frac{0.7}{3}+\frac{0.3}{4}+\frac{0.4}{5}$$

ii) $$A \cap B = \frac{0.5}{2}+\frac{0.5}{3}+\frac{0.2}{4}+\frac{0.2}{5}$$

iii) Complement of fuzzy set A : $1-\mu_A(x)$

$$= \left\{\frac{0}{2}+\frac{0.5}{3}+\frac{0.7}{4}+\frac{0.8}{5}\right\}$$

iv) Difference $\left(\frac{A}{B}\right) = A \cap \overline{B}$

$$= A-(A \cap B) = \left\{\frac{0.5}{2}+\frac{0}{3}+\frac{0.1}{4}+\frac{0}{5}\right\}$$

v) Algebraic sum $= \left\{\frac{1.5}{2}+\frac{1.2}{3}+\frac{0.5}{4}+\frac{0.6}{5}\right\}-\left\{\frac{0.5}{2}+\frac{0.35}{3}+\frac{0.06}{4}+\frac{0.08}{5}\right\}$

$$= \left\{\frac{1}{2}+\frac{0.85}{3}+\frac{0.44}{4}+\frac{0.52}{5}\right\}$$

vi) Bounded sum $= \min\left\{1, \left\{\frac{1.5}{2}+\frac{1.2}{3}+\frac{0.5}{4}+\frac{0.6}{5}\right\}\right\} = \left\{\frac{1.5}{2}+\frac{1.2}{3}+\frac{0.5}{4}+\frac{0.6}{5}\right\}$

vii) Algebraic product $= \left\{\frac{0.5}{2}+\frac{0.35}{3}+\frac{0.06}{4}+\frac{0.08}{5}\right\}$

viii) $$\overline{A \cup B} = 1-\max\{\mu_A(x), \mu_B(x)\}$$

$$= 1-\left\{\frac{1}{2}+\frac{0.7}{3}+\frac{0.3}{4}+\frac{0.4}{5}\right\} = \left\{\frac{0}{2}+\frac{0.3}{3}+\frac{0.7}{4}+\frac{0.6}{5}\right\}$$

ix) $A \cup \overline{B} = \max[\mu_A(x), \mu_B(x)]$

Calculate, $\overline{B} = 1-\mu_B(x)$

$$= 1-\left\{\frac{0.5}{2}+\frac{0.7}{3}+\frac{0.2}{4}+\frac{0.4}{5}\right\} = \left\{\frac{0.5}{2}+\frac{0.3}{3}+\frac{0.8}{4}+\frac{0.6}{5}\right\}$$

$$A \cup \overline{B} = \left\{\frac{1}{2}+\frac{0.5}{3}+\frac{0.8}{4}+\frac{0.6}{5}\right\}$$

Q.25 Consider two fuzzy sets X and Y, find Complement, Union, Intersection, Difference :

$$\mathbf{X} = \left\{\frac{0.5}{2}, \frac{0.4}{3}, \frac{0.1}{4}, \frac{0.1}{5}, \frac{0.3}{6}, \frac{0.7}{7}, \frac{0.8}{8}\right\}$$

$$\mathbf{Y} = \left\{\frac{0.3}{2}, \frac{0.8}{3}, \frac{0.6}{4}, \frac{0.8}{5}, \frac{0.2}{6}, \frac{0.2}{7}, \frac{0.3}{8}\right\}$$

Ans. : 1) Complement :

$$\text{Set X} = \left\{\frac{0.5}{2}, \frac{0.6}{3}, \frac{0.9}{4}, \frac{0.9}{5}, \frac{0.7}{7}, \frac{0.3}{7}, \frac{0.2}{8}\right\}$$

$$\text{Set Y} = \left\{\frac{0.7}{2}, \frac{0.2}{3}, \frac{0.4}{4}, \frac{0.2}{5}, \frac{0.8}{6}, \frac{0.8}{7}, \frac{0.7}{8}\right\}$$

2) Untion :

$$X \cup Y = \left\{\frac{0.5}{2}, \frac{0.8}{3}, \frac{0.6}{4}, \frac{0.8}{5}, \frac{0.3}{6}, \frac{0.7}{7}, \frac{0.8}{8}\right\}$$

3) Intersection :

$$X \cap Y = \left\{\frac{0.3}{2}, \frac{0.4}{3}, \frac{0.1}{4}, \frac{0.1}{5}, \frac{0.2}{6}, \frac{0.2}{7}, \frac{0.3}{8}\right\}$$

4) Difference

$$(X/Y) = X \cap \overline{Y} = X - (X \cap Y)$$

$$= \left\{\frac{0.2}{2}, \frac{0}{3}, \frac{0}{4}, \frac{0}{5}, \frac{0.1}{6}, \frac{0.5}{7}, \frac{0.5}{8}\right\}$$

Q.26 What do you mean by predicate logic ?

Ans. : A predicate is an expression of one or more variables defined on some specific domain. A predicate with variables can be made a proposition by either assigning a value to the variable or by quantifying the variable.

The following are some examples of predicates - Let E(x, y) denote "x = y

Q.27 What is fuzzy quantifiers ?

Ans. : Fuzzy quantifiers are fuzzy numbers that take part in fuzzy propositions.

- There are two different types: absolute and relative.

- Absolute quantifiers express quantities over the total number of elements of a particular set, stating whether this number is, for example, "much more than 10," "close to 100," "a great number of," and so forth.
- Relative quantifiers express measurements over the total number of elements, which fulfill a certain condition depending on the total number of possible elements.
- This type of quantifier is used in expressions such as "the majority" or "most," "the minority," "little of," "about half of," and so on

2.4 : Fuzzy Rule-Based Systems

Q.28 State fuzzy inference.

Ans. : Fuzzy inference can be defined as a process of mapping from a given input to an output, using the theory of fuzzy sets.

Q.29 What is a fuzzy rule ?

Ans. : A fuzzy rule can be defined as a conditional statement in the form :

IF x is A

THEN y is B

where x and y are linguistic variables; and A and B are linguistic values determined by fuzzy sets on the universe of discourses X and Y, respectively.

Q.30 What is fuzzy rule base system ?

Ans. : • A rule-based fuzzy logic system is comprised of four elements: rules, fuzzifier, inference engine and output processor.

- Fuzzy rules are linguistic IF-THEN - constructions that have the general form "IF A THEN B" where A and B are propositions containing linguistic variables.
- A is called the premise and B is the consequence of the rule. In effect, the use of linguistic variables and fuzzy IF-THEN- rules exploits the tolerance for imprecision and uncertainty. In this respect, fuzzy logic mimics the crucial ability of the human mind to summarize data and focus on decision-relevant information.
- In a more explicit form, if there are I rules each with K premises in a system, the i^{th} rule has the following form.

  If a_1 is $A_{i,1} \Theta a_2$ is $A_{i,2} \Theta \ldots \Theta a_k$ is $A_{i,k}$ then B_i.
- In the above equation a represents the crisp inputs to the rule and A and B are linguistic variables. The operator 1 can be AND or OR or XOR.

Q.31 What is linguistic variable ? Explain.

Ans. : • Linguistic variable is "a variable whose values are words or sentences in a natural or artificial language".

- Each linguistic variable may be assigned one or more linguistic values, which are in turn connected to a numeric value through the mechanism of membership functions.
- Conventional techniques for system analysis are intrinsically unsuited for dealing with systems based on human judgement, perception and emotion.
- A linguistic variable is characterized by a quintuple (x, T(x), X, G, M) in which x is the name of the variable, T(x) is the term set of x, X is the universe of discourse; G is syntactic rule which generates the terms in T(x) and M is a semantic rule which associates with each linguistic value A its meaning M(A), where M(A) denotes a fuzzy set in X.
- A fuzzy rule can be defined as a conditional statement in the form :

  If x is A THEN y is B

where x and y are linguistic variables; and A and B are linguistic values determined by fuzzy sets on the universe of discourses X and Y, respectively.

- Example : **If temperature is cold and oil is cheap then heating is high.**
- If **age is** interpreted as a linguistic variable, then its term **set T(age)** could be

  T(age) = { young, not young, very young,
  not very young, ...
  middle aged, not middle aged, ...
  old, not old, very old, more or
  less old, not very old, ...
  not very young and not very old, ...}

where each term is T(age) is characterized by a fuzzy set of a universe of discourse X = [0, 100].

- Here "age is young" is used to denote the assignment of the linguistic value "young" to the linguistic variable age.
- Let A be a linguistic value characterized by a fuzzy set with membership function $\mu_A(.)$. Then A^k is interpreted as a modified version of the original linguistic value expressed as

$$A^k = x[\mu_A(x)]^k / x$$

Q.32 Explain fuzzy IF-Then rules.

Ans. : • Fuzzy sets and fuzzy sets operations are the subjects and verbs of fuzzy logic. If-Then rule statements are used to formulate the conditional statements that comprise fuzzy logic.

- A fuzzy if-them rule is also called fuzzy rule, fuzzy implication or fuzzy conditional statement. It assume the form

If x is A then y is B

where A and B are linguistic values defined by fuzzy sets on universes of discourse X and Y respectively.

- Often "x is A" is called the **antecedent or premise** while "y is B" is called the **consequence or conclusion.**
- The antecedent describes to what degree the rule applies, while the conclusion assigns a fuzzy function to each of one or more output variables.

Examples :

- If pressure is high, then volume is small.
- If the road is slippery, then driving is dangerous.
- If a tomato is red, then it is ripe.
- If the speed is high, then apply the brake a little.

Q.33 What do you mean defuzzification ?

Ans. : The last step in the fuzzy inference process is defuzzification. Fuzziness helps us to evaluate the rules, but the final output of a fuzzy system has to be a crisp number. The input for the defuzzification process is the aggregate output fuzzy set and the output is a single number.

Q.34 What is defuzzification ? Explain different methods of defuzzification.

Ans. : • Defuzzification means the fuzzy-to-crisp conversions. The fuzzy results generated cannot be used to the applications. It is necessary to convert the fuzzy quantities into crisp quantities for further processing.

- Defuzzification has the capability to reduce a fuzzy set to a crisp single-valued quantity or as a set.
- Defuzzification is the process by which a fuzzy consequent is reduced to a singleton or crisp scalar value in order to provide an interface to a typically scalar 'action'. Actions are typically associated with a scalar defining a specific measurable property.
- The task of defuzzification is to find one single crisp value that summarizes the fuzzy set that enters it from the inference block. Examples might include speed of movement, volume of voice, (angular) direction, or a decrease/increase in health/energy.
- Defuzzification operates on the implied fuzzy sets produced by the inference mechanism and combines their effects to provide the "most certain" controller output.
- Defuzzification methods are as follows :
  1. Maxima method
  2. Centroid method
  3. Weighted average method
  4. Middle-of-maxima method
  5. First-of-maxima or last-of-maxima

Max Membership Principle

- For peaked output function, max membership is used. It is also called as height method. Fig. Q.34.1. shows maximum membership principle.

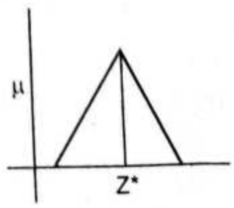


Fig. Q.34.1

- The method is expressed as :

$$\mu_c(Z^*) \geq \mu_c(z)$$

Where Z* is the defuzzified value.

Centroid Method

- In the centroid method, the crisp value of the output variable is computed by finding the variable value of the centre of gravity of the membership function for the fuzzy value. Fig. Q.34.2 shows centroid method.

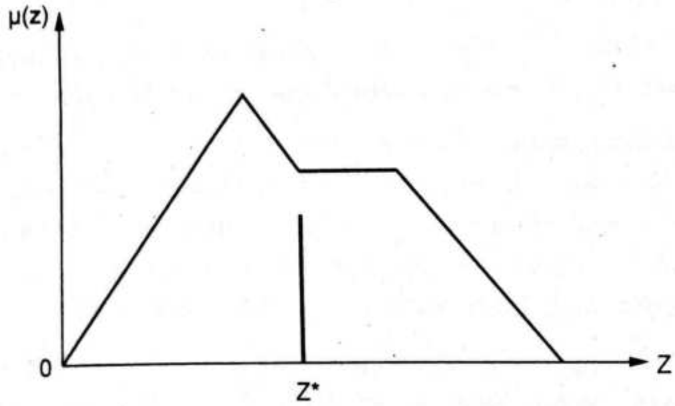


Fig. Q.34.2 Centroid method

- This most familiar defuzzification scheme. The composite output fuzzy set is built by taking the union of all clipped or scaled output fuzzy sets.
- The crisp output value is then obtained by deriving the centroid or centre of mass of the shape represented by the composite output fuzzy set.
- The x co-ordinate of the centre of mass is the required crisp output value. Because of the variety of shapes that are encountered, the method is computationally intensive.

$$z^* = \frac{\int \mu_c(z).z\,dz}{\int \mu_c(z)\,dz}$$

where $\int$ denotes an algebraic integration.

Weighted Average Method

- Weighted average defuzzification instead of using a composite output membership function, uses the individual clipped or scaled output fuzzy sets.
- The method takes the peak value of each clipped/scaled output fuzzy set and builds the weighted (with respect to the peak heights) sum of these peak values. The output is given by :

$$z^* = \frac{\sum \mu_{\underset{\sim}{C}}(\bar{z})\,\bar{z}}{\sum \mu_{\underset{\sim}{C}}(\bar{z})}$$

$$z^* = \frac{a(0.8)+b(0.6)}{0.8+0.6}$$

$$z^* = \frac{0.8a+0.6b}{1.4}$$

- The weighted average method is formed by weighting each membership function in the output by its respective maximum membership value.

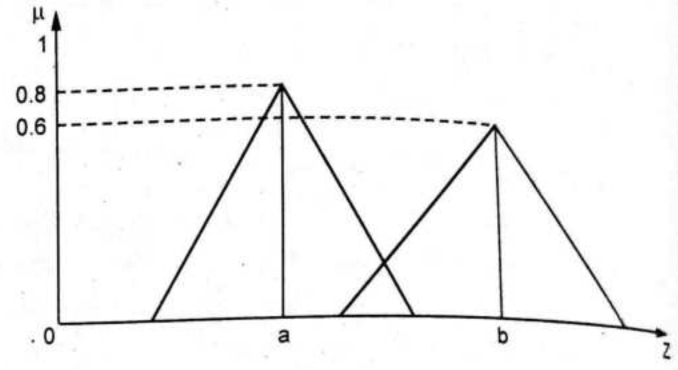


Fig. Q.34.3 Weighted average method

- This method can be used only for symmetrical output membership functions.

Other Methods

a. Centre of sums

- As with the centroid method, the clipped or scaled output fuzzy sets are combined to form a composite output fuzzy set. The contribution of each clipped or scaled fuzzy set, however, is considered individually.
- Unlike the Centroid method which builds the output composite set using the union operation the centre of sums method takes the SUM of the clipped/scaled output Fuzzy sets and then computes the centre of mass of the resulting shape.
- Thus overlapping areas, if they exist are reflected more than once by this technique.

b. Singleton method

- The singleton method is quite a simple method that is widely used in hardware implementations. The technique uses output membership functions which are just vertical lines of height 1 in output space, or singletons.
- Each singleton has a single crisp output value associated with it. The final crisp output is a weighted average of all qualified output sets.

Q.35 Explain one application of fuzzy system.

Ans. : • Aircraft landing is the process during which the aircraft descends from certain altitude and touches the ground. The aircraft landing system is supposed to control the vertical speed during the aircraft landing process, so that the aircraft touches the ground "smoothly" but descends rapidly.

- If the aircraft touches the ground with high negative vertical speed, that is considered a crash landing and should be avoided by the controller. Fig. Q.35.1 shows landing problem of aircraft.

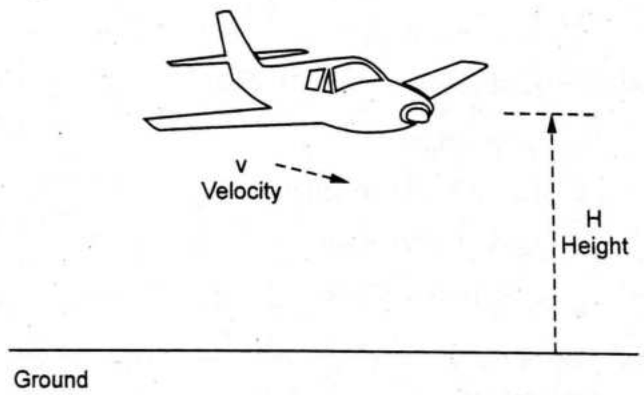


Fig. Q.35.1 Landing problem of aircraft

- Vertical velocity changes only if control force is applied. Negative control force directs the aircraft to the ground while positive control force directs the aircraft from the ground.

1. **Antecedent :** Height "h" above ground and vertical velocity of the aircraft, "v"
2. **Consequent :** Control forces that, when applied to the aircraft, will alter its height, h and velocity, v.
3. **Membership functions :**
   a. **Height :** L (large), M (medium), S (small), NZ (near zero)
   b. **Velocity :** UL(up large), US (up small), Z (zero), DS(down small), DL (down large)
   c. **Force :** UL (up large), US (up small), Z (zero), DS (down small), DL (down large)

- If current vertical velocity is negative, than aircraft will descend, if it is positive it will increase altitude and if it is zero, then aircraft will remain at the same altitude.
- All three different types of controllers have two inputs : Height in feet and vertical velocity in feet per second.
- The output is also the same variable for all three types of controllers : Control force in pound-force. Control force can change current vertical velocity, and change of vertical velocity, changes rate of descent or climb, thus affects current height.
- Fig. Q.35.2 shows fuzzy aircraft landing control system.

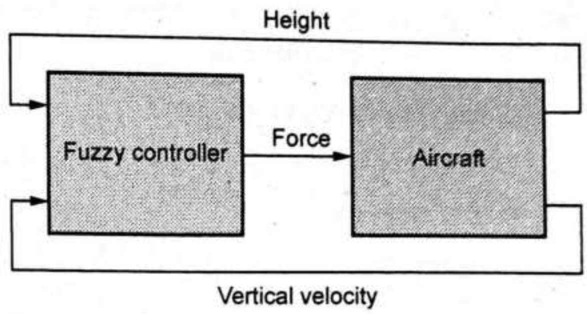


Fig. Q.35.2 Fuzzy aircraft landing control system

- Fuzzy rules are as follows :

		Velocity				
		DL	DS	ZERO	US	UL
Height	L	Z	DS	DL	DL	DL
	M	US	Z	DS	DL	DL
	S	UL	US	Z	DS	DL
	NZ	UL	UL	Z	DS	DS

- Control surface graphically represents all possible inputs and outputs.

Fill in the Blanks for Mid Term Exam

Q.1 A fuzzy set is a set having degree of membership between _____ and _____ .

Q.2 Fuzzy logic is a form of _____ logic.

Q.3 A fuzzy graph is a _____ representation of a binary fuzzy relation.

Q.4 Fuzzy logic is usually represented as _____ rules.

Q.5 _____ is the process of transforming a crisp set to a fuzzy set to a fuzzier set.

Q.6 _____ is the process of conversion of a fuzzy quantity into a precise quantity.

Q.7 Fuzzy logic uses _____ variables.

Multiple Choice Questions for Mid Term Exam

Q.1 Fuzzy logic is a form of ___________.

a two-valued logic
b crisp set logic
c many-valued logic
d binary set logic

Q.2 How is Fuzzy Logic different from conventional control methods ?

a IF and THEN approach
b FOR approach
c WHILE approach
d DO approach

Q.3 Both fuzzy logic and artificial neural network are soft computing techniques because _____.

a both gives precise and accurate results.
b artificial neural network gives accurate result, but fuzzy logic does not.
c in each, no precise mathematical model of the problem is required.
d fuzzy gives exact result but artificial neural network does not.

Q.4 Which of the following cannot be stated using fuzzy logic ?

a Color of an apple
b Height of a person
c Date of birth of a student
d Speed of a car

Q.5 The height h(A) of a fuzzy set A is defined as h(A) = support A(x), where x belongs to A. Then the fuzzy set A is called normal when

a $h(A) = 0$
b $h(A) < 0$
c $h(A) = 1$
d $h(A) > 1$

Q.6 The cardinality of the given set A = {1, 2, 3, 4, 5}

a 2
b 5
c 4
d 1

Q.7 If X is A then Y is B else Y is C. The output of the given fuzzy rule is _______.

a a fuzzy set
b a crisp set
c a membership function
d a fuzzy relation

Q.8 The cardinality of the fuzzy set on any universe is ___________.

a infinity
b 0
c 1
d − 1

Q.9 Defuzzification is done to obtain _________.

a crisp output
b the best rule to follow
c precise fuzzy value
d none of the above

Q.10 "The train is running fast". Here 'fast' can be represented by ___________.

a fuzzy Set
b crisp Set
c fuzzy & Crisp Set
d none of the mentioned

Answer Keys for Fill in the Blanks :

Q.1	1, 0	Q.2	many-valued
Q.3	graphical	Q.4	IF-THEN
Q.5	Fuzzification	Q.6	Defuzzification
Q.7	linguistic		

Answer Keys for Multiple Choice Questions :

Q.1	c	Q.2	a
Q.3	c	Q.4	c
Q.5	c	Q.6	b
Q.7	d	Q.8	a
Q.9	a	Q.10	a

END...

UNIT - III

3 Fuzzy Decision Making

3.1 : Fuzzy Decision Making

Q.1 What is fuzzy decision making ?

Ans. : Fuzzy decision making is the collection of single or multicriteria techniques aiming at selecting the best alternative in case of imprecise, incomplete, and vague data.

Q.2 List the various types of fuzzy decision making.

Ans. : Various kind of fuzzy decision making are

a. Individual decision making
b. Multi-person decision making
c. Multi-object decision making
d. Multi-attribute decision making
e. Fuzzy Bayesian decision making

Q.3 What are steps for decision making ?

Ans. : The steps involved in the decision-making process are as follows :

1. Determining the set of alternatives : In this step, the alternatives from which the decision has to be taken must be determined.
2. Evaluating alternative : the alternatives must be evaluated so that the decision can be taken about one of the alternatives.
3. Comparison between alternatives : In this step, a comparison between the evaluated alternatives is done

Q.4 Explain following decision making :
a. Individual decision making
b. Multi-person decision making

Ans. : a) Individual decision making :

- Only a single person is responsible for taking decisions.
- A decision situation in this model is characterized by :
  i. Set of possible actions
  ii. Set of goals $P_i (i \in x_n)$, expressed in terms of fuzzy set
  iii. Set of constraints $Q_i (j \in x_m)$, expressed in terms of fuzzy sets.

b) Multi-person decision making :

- When decision are made by many persons, the difference of it from the individual decision maker is :
  1. The goals of single decision makers differ, such that each places a different ordering arrangements.
  2. The individual decision makers have access to different information upon which to base their decision.
- In this case, each member of a group of n single decision makers has a preference ordering P_k, $K \in N_n$, which totally or partially orders a set x.

3.2 : Particle Swarm Optimization

Q.5 What is an agent ? What is a Swarm ? Define swarm intelligence. Give an examples of Swarms in nature.

Ans. : • An agent is anything that can be viewed as perceiving its environment through sensors and acting upon that environment through effectors.

- Swarm is a loosely structured collection of interacting agents. Agents is an individuals that belong to a group. They contribute to and benefit from the group. They can recognize, communicate, and/or interact with each other.

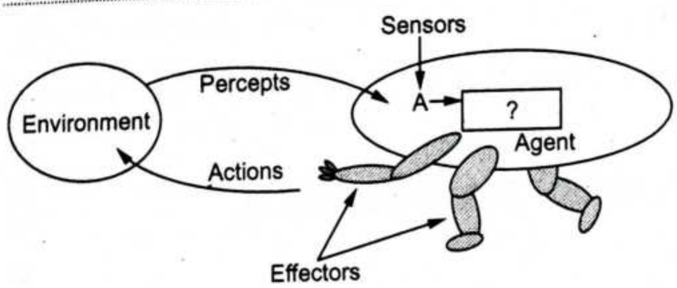


Fig. Q.5.1

- The instinctive perception of swarms is a group of agents in motion, but that does not always have to be the case.
- A swarm is better understood if thought of as agents exhibiting a collective behavior.
- Examples of Swarms in Nature :
  1. Classic Example : Swarm of Bees
  2. Can be extended to other similar systems : Ant colony (Agents : ants), Flock of birds(Agents : birds), Traffic(Agents : cars) , Crowd(Agents : humans), Immune system(Agents : cells and molecules)
- Swarm Intelligence : Any attempt to design algorithms or distributed problem-solving devices inspired by the collective behavior of social insect colonies and other animal societies.
- Swarm intelligence (SI) is an artificial intelligence technique based around the study of collective behavior in decentralized, self-organized systems

Q.6 List the advantages of swarm intelligence.

Ans. : • The systems are scalable because the same control architecture can be applied to a couple of agents or thousands of agents.

- The systems are flexible because agents can be easily added or removed without influencing the structure.
- The systems are robust because agents are simple in design, the reliance on individual agents is small, and failure of a single agents has little impact on the system's performance.
- The systems are able to adapt to new situations easily.

3.3 : Particle Swarm Optimization (PSO) Algorithm

Q.7 Write short note on particle swarm optimization

Ans. : • Particle Swarm Optimization (PSO) is a population based stochastic optimization technique developed by Dr. Eberhart and Dr. Kennedy in 1995, inspired by social behavior of bird flocking or fish schooling.

- PSO system combines local search methods with global search methods, attempting to balance exploration and exploitation.
- Suppose a group of birds is searching food in an area. Only one piece of food is available. Birds do not have any knowledge about the location of the food. But they know how far the food is from their present location.
- **So what is the best strategy to locate the food ? The best strategy is to follow the bird nearest to the food.**
- A flying bird has a position and a velocity at any time. In search of food, the bird changes his position by adjusting the velocity. The velocity changes based on his past experience and also the feedbacks received from his neighbor.
- This searching process can be artificially simulated for solving non-linear optimization problem. So this is a population based stochastic optimization technique inspired by social behaviour of bird flocking or fish schooling.
- PSO shares many similarities with evolutionary computation techniques such as Genetic Algorithms. The system is initialized with a population of random solutions and searches for optima by updating generations.
- In PSO algorithms, the population $P = \{p_1,\ldots,p_n\}$ of the feasible solutions is often called a **swarm.** The feasible solutions $p_1,\ldots,p_n$ are called **particles.**
- However, unlike GA, PSO has no evolution operators such as crossover and mutation. In PSO, the potential solutions, called particles, fly through the problem space by following the current optimum particles.

- Each particle keeps track of its coordinates in the problem space which are associated with the best solution (fitness) it has achieved so far.
- The fitness value is also stored. This value is called pbest. Another "best" value that is tracked by the particle swarm optimizer is the best value, obtained so far by any particle in the neighbors of the particle. This location is called lbest. When a particle takes all the population as its topological neighbors, the best value is a global best and is called gbest.
- The particle swarm optimization concept consists of, at each time step, changing the velocity of (accelerating) each particle toward its pbest and lbest locations. Acceleration is weighted by a random term, with separate random numbers being generated for acceleration toward pbest and lbest locations.
- In past several years, PSO has been successfully applied in many research and application areas. It is demonstrated that PSO gets better results in a faster, cheaper way compared with other methods.
- **Advantages and disadvantages**
  1. The fitness function can be non-differentiable
  2. There is no general convergence theory applicable to practical, multidimensional problems.

Q.8 Compare Particle Swarm Optimization (PSO) with Genetic Algorithms (GA).

Ans. : • Unlike GAs, PSOs do not change the population from generation to generation, but keep the same population, iteratively updating the positions of the members of the population (i.e., particles).

- PSOs have no operators of "mutation", "recombination", and no notion of the "survival of the fittest".
- On the other hand, similarly to GAs, an important element of PSOs is that the members of the population "interact", or "influence" each other.
- GA usually converges towards a local optimum or even arbitrary points rather than the global optimum of the problem while as PSO tries to find the global optima.
- GA is discrete in nature, i.e. it changes the variables into binary 0's and 1's, and therefore it can easily handle discrete problems, and PSO is continuous and hence must be modified in order to handle discrete problems.

Q.9 Write an algorithm of particle swarm optimization.

Algorithm parameters :

A	:	Population of agents
$\mathbf{p_i}$	:	Position of agent a_i in the solution space
f	:	Objective function
$\mathbf{v_i}$	:	Velocity of agent's a_i
$\mathbf{V(a_i)}$	:	Neighborhood of agent a_i (fixed)

- Particle update rule : $p = p + v$

With $v = v + c_1^* \text{ rand} * (\text{pBest} - p) + c_2^* \text{ rand} * (\text{gBest} - p)$

where

p	:	Particle's position
v	:	Path direction
c_1	:	Weight of local information
c_2	:	Weight of global information
pBest	:	Best position of the particle
gBest	:	Best position of the swarm
rand	:	Random variable

Algorithm :

1. Create a 'population' of agents (particles) uniformly distributed over X.
2. Evaluate each particle's position according to the objective function.
3. If a particle's current position is better than its previous best position, update it.
4. Determine the best particle (according to the particle's previous best positions).
5. Update particles' velocities :

$$v_i^{t+1} = \underbrace{v_i^t}_{\text{inertia}} + \underbrace{c_1 U_1^t (pb_i^t - p_i^t)}_{\text{personal influence}} + \underbrace{c_2 U_2^t (gb^t - p_i^t)}_{\text{social inflence}}$$

Ring Topology :

- Fig. Q.12.2 shows ring topology.
- lbest model : Each particle is influenced only by particles in its own neighbourhood.
- The propagation of information is the slowest.
- Doesn't get stuck easily in local minima but might increase the computational cost.
- Note that, neighborhood topologies are usually defined by the particle index, and not by the particle location.

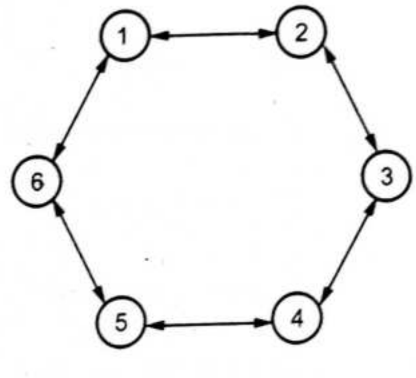


Fig. Q.12.2

Q.13 Explain global best (gbest) and local best (lbest) particle swarm optimization.

Ans. : Global best (gbest) PSO :

- For the global best (gbest) PSO, the neighborhood for each particle is the entire swarm. The social network employed by the gbest PSO reflects the star topology.
- gbest : each particle is influenced by the best found from the entire swarm.
- Velocity update per dimension :

$$v_{ij}(t+1) = v_{ij}(t) + c_1 r_{1j}(t)[y_{ij}(t) - x_{ij}(t)] + c_2 r_{2j}(t)[\hat{y}_j(t) - x_{ij}(t)]$$

$v_{ij}(0) = 0$ (usually, but can be random) and c_1, c_2 are positive acceleration coefficients.

gbest PSO Algorithm :

```
Create and initialize an n_x-dimensional swarm, S;
 repeat
     for each particle i = 1, . . . , S.n_s do
      if f(S·y_i) < f(S·y) then
        S·y_i = S·x_i ;
       end
      if f(S·y_i) < f(S·^y) then
          S·^y = S·y_i ;
      end
   end
for each particle i = 1, . . . , S·n_s do
update the velocity and then the position;
end
until stopping condition is true ;
```

Local best PSO

- The local best (lbest) PSO, uses a ring social network topology where smaller neighborhoods are defined for each particle.
- The social component reflects information exchanged within the neighborhood of the particle, reflecting local knowledge of the environment.

- With reference to the velocity equation, the social contribution to particle velocity is proportional to the distance between a particle and the best position found by the neighborhood of particles.
- Due to the larger particle interconnectivity of the gbest PSO, it converges faster than the lbest PSO. However, this faster convergence comes at the cost of less diversity than the lbest PSO.
- The lbest model is less vulnerable to the attraction of local optima but with a slower convergence speed than the gbest model.

lbest PSO algorithm :

```
Create and initialize and n_x-dimensiional swarm;
  repeat
     for each particle i = 1 ... n_s do
        //set the personal best position
        if f(x_i) < f (y_i) then
           y_i = x_i;
        end
        //set the neighborhood best position
        if f (y_i) < f (ŷ_i) then
           ŷ = y_i;
     end
  end
  for each particle i = 1,...n_s do
     update the velocity
     update the position
  end
until stopping condition is true;
```

Q.14 Why used PSO ?

Ans. :

- With a population of candidate solutions, a PSO algorithm can maintain useful information about characteristics of the environment.
- PSO, as characterized by its fast convergence behavior, has an in-built ability to adapt to a changing environment.
- Some early works on PSO have shown that PSO is effective for locating and tracking optima in both static and dynamic environments.

Q.15 Explain difference between gbest and lbest PSO.

gbest PSO	lbest PSO
Global best particle swarm optimization uses a "star" neighborhood topology where all the particles communicate with each other.	Local best particle swarm optimization can be used with different neighborhood topologies with the "ring" topology being the most common.
gbest PSO converges faster compared with local best particle swarm optimization.	lbest PSO converges is slower compared with global best particle swarm optimization.
For the global best PSO, the neighborhood for each particle is the entire swarm.	In lbest PSO, smaller neighborhoods are defined for each particle.
The gbest PSO performs best for unimodal problems.	In lbest, ring structure to provide better performance in terms of the quality of solutions found for multi-modal problems than the star structure.

Fill in the Blanks for Mid Term Exam

Q.1 Swarm is a loosely structured collection of interacting ________.

Q.2 Particle swarm optimization is a ________ based stochastic optimization technique

Q.3 When a particle takes all the population as its topological neighbors, the best value is a global best and is called ________.

Q.4 In BPSO, each solution in the population is a ________ string.

Q.5 The social network employed by the gbest PSO reflects the ________ topology

Q.6 The ________ best PSO, uses a ring social network topology where smaller neighbourhoods are defined for each particle.

Multiple Choice Questions for Mid Term Exam

Q.1 Various kind of fuzzy decision making are ______.

a individual
b multi-person
c multi-object
d all of these

Q.2 Particle swarm optimization is a ______ stochastic approach for solving continuous and discrete optimization problems.

a random
b population-based
c machine learning
d fuzzy based

Q.3 In PSO algorithms, the population P = {p1,...,pn} of the feasible solutions is often called a ______.

a particles
b agent
c swarm
d all of these

Q.4 The ______ PSO, uses a ring social network topology where smaller neighborhoods are defined for each particle.

a global best
b local best
c distributed best
d remote best

Q.5 Global best particle swarm optimization uses a ______ neighborhood topology where all the particles communicate with each other.

a mesh
b ring
c star
d bus

Answer keys for Fill in the Blanks :

Q.1	agents	Q.2	population
Q.3	gbest	Q.4	binary
Q.5	star	Q.6	local

Answer keys for Multiple Choice Questions :

Q.1	d	Q.2	b
Q.3	c	Q.4	b
Q.5	c		

END...

UNIT - IV

4 Genetic Algorithm

4.1 : Basic Concepts of Genetic Algorithm

Q.1 What is genetic algorithm ?

Ans. : A genetic algorithm is a search technique used in computing to find true or approximate solutions to optimization and search problems.

Q.2 What do you mean genetic programming ?

Ans. : Genetic programming is genetic algorithm wherein the population contains programs rather than bit strings.

Q.3 Explain the "Darwinian theory of survival".

Ans. : More individuals are produced each generation that can survive. Phenotypic variation exists among individuals and the variation is heritable. Those individuals with heritable traits better suited to the environment will survive.

Q.4 State the importance of genetic algorithm.

Ans. : Genetic algorithm is that the problem solving strategy involves using "the strings' fitness to direct the search; therefore they do not require any problem-specific knowledge of the search space, and they can operate well on search spaces that have gaps, jumps, or noise".

- As each individual string within a population directs the search, the genetic algorithm searches, in parallel, numerous points on the problem state space with numerous search directions.

Q.5 Compare and contract genetic algorithm with traditional algorithm.

Ans. :

Genetic algorithm	Traditional algorithm
GA generates a population of points at each iteration. The best point in the population approaches an optimal solution.	It generates a single point at each iteration. The sequence of points approaches an optimal solution.
Selects the next population by computation which uses random number generators	Selects the next point in the sequence by a deterministic computation.
Convergence in each iteration in problem independent	Improvement in each iteration is problem specific
Rules are probabilistic.	Rules are fully deterministic.

Q.6 What two requirements should a problem satisfy it by a genetic algorithm ?

Ans. : GA can only be applied to problems that satisfy the following requirements :

i) These function can be well defined.

ii) Solutions should be decomposable into steps (building blocks) which could be then encoded as chromosomes

Q.7 List the basic components used in all genetic algorithms.

Ans. : The basic components common to almost all genetic algorithms are :

1. Fitness function for optimization
2. a population of chromosomes
3. selection of which chromosomes will reproduce
4. crossover to produce next generation of chromosomes
5. random mutation of chromosomes in new generation

Q.8 Why use Genetic algorithm ?

Ans. : • Genetic algorithms evaluate the target function to be optimized at some randomly selected points of the definition domain. Genetic algorithms can be used when no information is available about the gradient of the function at the evaluated points. The function itself does not need to be continuous or differentiable.

- Genetic algorithms can still achieve good results even in cases in which the function has several local minima or maxima.
- Genetic algorithms are stochastic search algorithms which act on a population of possible solutions. They are loosely based on the mechanics of population genetics and selection.
- Genetic algorithms allow you to explore a space of parameters to find solutions that score well according to a "fitness function".
- Genetic programming takes genetic algorithms a step further, and treats programs as the parameters. For example, you would breeding path finding algorithms instead of paths, and your fitness function would rate each algorithm based on how well it does.

Q.9 Define : a. Cell b. Genotype c. Phenotype

Ans. : a. Cell : Animal or human cell is a complex of many small factories that work together. The center of all this is the cell nucleus. The genetic information is contained in the cell nucleus.

b. Genotype for a particular individual, the entire combination of genes is called genotype.

c. Phenotype the phenotype describes the physical aspect of decoding a genotype to produce the phenotype.

Q.10 What is Chromosome ? Explain.

Ans. : • All living organisms consist of cells. In each cell there is the same set of chromosomes. Chromosomes are strings of DNA and serves as a model for the whole organism.

- A chromosome consists of genes, blocks of DNA. Each gene encodes a particular protein.
- Basically can be said, that each gene encodes a trait, for example color of eyes. Possible settings for a trait (e.g. blue, brown) are called alleles.
- Each gene has its own position in the chromosome. This position is called locus.
- Complete set of genetic material (all chromosomes) is called genome. Particular set of genes in genome is called genotype.
- The genotype is with later development after birth base for the organism's phenotype, its physical and mental characteristics, such as eye color, intelligence etc

Q.11 Describe working principle of genetic algorithm.

Ans. : 1. **[Start]** Generate random population of n chromosomes (suitable solutions for the problem).

2. **[Fitness]** Evaluate the fitness **f(x)** of each chromosome **x** in the population.
3. **[New population]** Create a new population by repeating following steps until the new population is complete.
   a. **[Selection]** Select two parent chromosomes from a population according to their fitness (the better fitness, the bigger chance to be selected).
   b. **[Crossover]** with a crossover probability cross over the parents to form a new offspring (children). If no crossover was performed, offspring is an exact copy of parents.
   c. **[Mutation]** with a mutation probability mutate new offspring at each locus (position in chromosome).
   d. **[Accepting]** Place new offspring in a new population.
4. **[Replace]** Use new generated population for a further run of algorithm.
5. **[Test]** If the end condition is satisfied, stop and return the best solution in current population.
6. **[Loop]** Go to step **2**.

GA Steps

Step 1 : Set population size and probability

Step 2 : Define fitness function

Step 3 : Generate initial population

Step 4 : Calculate fitness for each chromosome

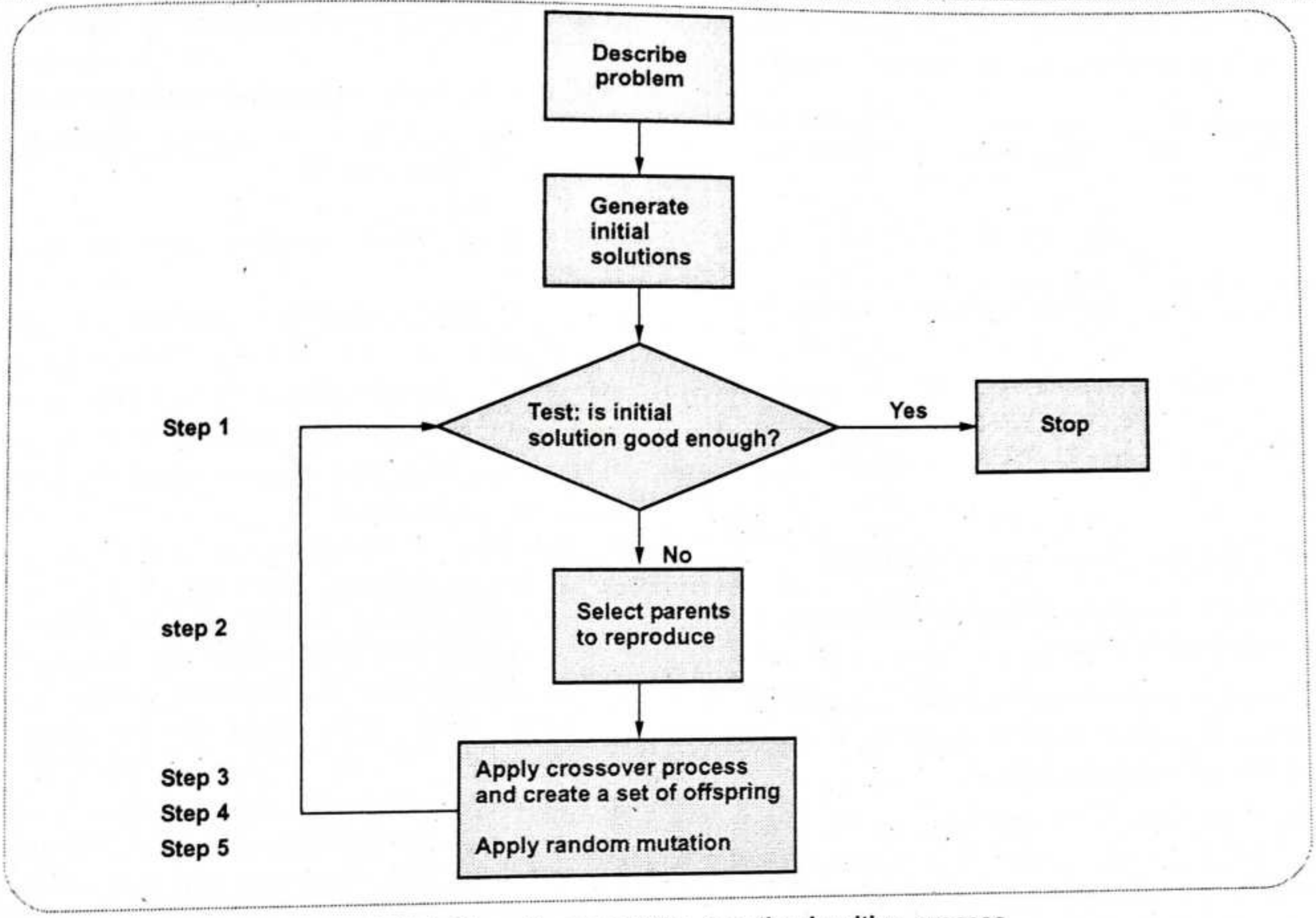


Fig. Q.11.1 Flow diagram of the genetic algorithm process

Step 5 : Mating of chromosomes

Step 6 : Create offspring-crossover and mutation

Step 7 : Offspring in new population

Step 8 : Repeat Step 5 until new = initial population

Step 9 : Replace initial population with new

Step 10 : To Step 4 and repeat until criteria achieved

Q.12 How does genetic algorithm differ from conventional algorithm ? Give the advantages of GA over conventional algorithms.

Ans. : 1. Genetic algorithms a coded form of the function values i.e. parameter set, rather than with the actual values themselves. For example, if we want to find the minimum of the function $f(x) = x^3 + x^2 + 5$, the GA would not deal directly with x or y values, but with strings that encode these values. For this case, strings representing the binary x values should be used.

2. Genetic algorithms use a set or population of points to conduct a search, not just a single point on the problem space. This gives GAs the power to search noisy spaces littered with local optimum points. Instead of relying on a single point to search through the space, the GAs looks at many different areas of the problem space at once, and uses all of this information to guide it.

3. Genetic algorithms use only payoff information to guide themselves through the problem space. Many search techniques need a variety of information to guide themselves. Hill climbing methods require derivatives, for example. The only information a GA needs is some measure of fitness about a point in the space. Once the GA knows the current measure of "goodness" about a point, it can use this to continue searching for the optimum.

4. GAs are probabilistic in nature, not deterministic. This is a direct result of the randomization techniques used by GAs.

5. GA is inherently parallel. Here lies one of the most powerful features of genetic algorithms.

GA's, by their nature, are very parallel, dealing with a large number of points simultaneously.

Parameters	Genetic Algorithms	Traditional Methods
Work with	Coding of parameter set	Parameters directly
Use information	Payoff i.e. objective function	Payoff plus derivatives etc.
Rules	Probabilistic	Fully deterministic
Search	A population of points	A population of points a single point

- Genetic algorithm is a stochastic algorithm.
- Randomness as an essential role in both selection and reproduction phases.
- Genetic algorithms always consider a population of solutions. A population base algorithm is also very amenable for parallelization.
- There is no particular requirement on the problem before using genetic algorithms, so it can be applied to resolve any problem (optimization).
- GAs are a new field and parts of the theory have still to be properly established. We can find almost as many opinions on GAs as there are researchers in this field.

Q.13 What are limitations on genetic algorithms ?

Ans. : • GAs are not guaranteed to find the global optimum solution to a problem.

- GAs are an extremely general tool and they have no specific way for solving particular problems.
- GAs are usually used when everything else is failed or when we don't have enough knowledge of the search space.
- Even when such specialized techniques exist, it is often interesting to hybridise them with a GA in order to possibly gain some improvements.

4.2 : Basic Operators for Genetic Algorithms

Q.14 List the basic operators of genetic algorithm.

Ans. : Basic operators of genetic algorithm are encoding, selection, mutation.

Q.15 Explain basic operator used in Genetic algorithm.

Ans. : • A genetic algorithm maintains a population of candidate solutions for the problem at hand and makes it evolve by iteratively applying a set of stochastic operators. The simplest form of genetic algorithm involves three types of operators : selection, crossover and mutation.

1. **Selection** : Selection replicates the most successful solutions found in a population at a rate proportional to their relative quality. This operator selects chromosomes in the population for reproduction. The fitter the chromosome, the more times it is likely to be selected to reproduce.
2. **Crossover** : This operator randomly chooses a locus and exchanges the subsequences before and after that locus between two chromosomes to create two offspring. For example, the strings 10000100 and 11111111 could be crossed over after the third locus in each to produce the two offspring 10011111 and 11100100. The crossover operator roughly mimics biological recombination between two single chromosome (haploid) organisms.
3. **Mutation** : Mutation randomly perturbs a candidate solution. This operator randomly flips some of the bits in a chromosome. For example, the string 00000100 might be mutated in its second position to yield 01000100. Mutation can occur at each bit position in a string with some probability, usually very small (e.g., 0.001).

Q.16 Reproduction is sometime called as selection operator. Why ?

Ans. : Reproduction selects good strings in a population and forms a mating pool. This is one of the reasons for the reproduction operation to be sometimes known as the selection operator.

Q.17 What are the various methods of selecting chromosomes for parents to crossover ?

Ans. : Various methods are :

1. Roulette-Wheel selection 2. Boltzmann selection

3. Tournament selection 4. Rank selection

5. Steady state selection

Q.18 How is a population with increasing fitness generated ?

Ans. : If two parents have superior fitness, there is a good chance that a combination of their genes will produce an offspring with even higher fitness.

Q.19 Define mutation rate.

Ans. : Mutation rate is the probability of mutation which is used to calculate number of bits to be muted.

Q.20 Explain different types of mutation function in Genetic algorithm ?

Ans. : Following are the Types of Mutation function in GA :

1. Insert Mutation- It is used in Permutation encoding.
2. Inversion Mutation- Inversion mutation is used for chromosomes with permutation encoding.
3. Scramble Mutation- Scramble mutation is also used with permutation encoded chromosome.
4. Swap Mutation- It is also used in Permutation encoding
5. Flip Mutation- Based on a generated mutation chromosome, flipping of a bit involves changing 0 to 1 and 1 to 0.
6. Interchanging Mutation - Two random positions of the string are chosen and the bits corresponding to those positions are interchanged.
7. Reversing Mutation - In case of reversing mutation applied for binary encoded chromosome, random position is chosen and the bits next to that position are reversed and child chromosome is produced.
8. Uniform Mutation - The Mutation operator changes the value of chosen gene with uniform random value selected between the user specified upper and lower bound for that gene. It is used in case of real and integer representation.
9. Creep Mutation - In creep mutation, a random gene is selected and its value is changed with a random value between lower and upper bound. It is used in case of real representation

Q.21 What is cross-over ? Explain various types of cross-over operations used in GA.

Ans. : • A binary variation operator is called recombination or crossover. A method of mixing good solutions to produce better ones is called crossover.

- Crossover means choosing a random position in the string (say, after 2 digits) and exchanging the segments either to the right or to the left of this point with another string partitioned similarly to produce two new offspring.
- Crossover produces two new offspring from two parent strings by copying selected bits from each parent. Bit at position "i" in each offspring is copied from the bit at position "I" in one of the two parents. The choice which parent contributes bit "I" is determined by an additional string, called **cross-over mask**.
- In the crossover operator, new strings are created by exchanging information among strings of the mating pool.
- The primary objective of the recombination operator is to emphasize the good solutions and eliminate the bad solutions in a population, while keeping the population size constant.
- "Selects The Best, Discards The Rest". "Recombination" is different from "Reproduction".
- Identify the good solutions in a population. Make multiple copies of the good solutions. Eliminate bad solutions from the population so that multiple solutions from the population so that multiple copies of good solutions can be placed in the population. It is the process in which two chromosomes (strings) combine their genetic material (bits) to produce a new offspring which possesses both their characteristics. Two strings are picked from the mating pool at random to crossover. The method chosen depends on the encoding method.

- Cross over can be rated complicated and very depends on encoding of the chromosome. Specific crossover made for a specific problem can improve performance of the genetic algorithm.

Steps of crossover :

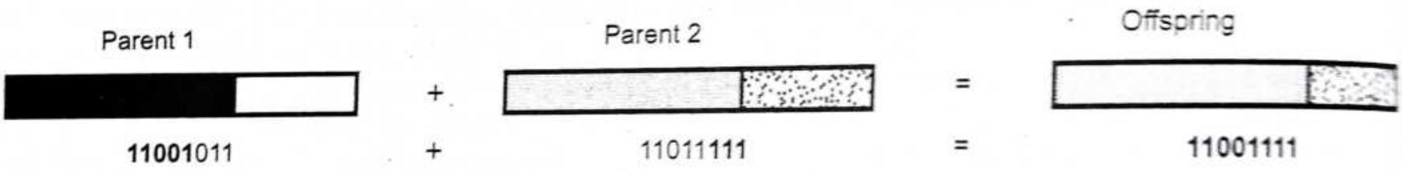
1. The reproduction operator selects at random a pair of two individual strings for the mating.
2. A cross site is selected at random along the string length.
3. Finally, the position values are swapped between the two strings following the cross site.

Crossover Types

1. Single-point crossover,
2. Two-point crossover,
3. Uniform crossover

One point crossover

- One point crossover is the most basic crossover operator. Crossover point on the genetic code is selected at random and two parent chromosomes are interchanged at this point. Binary string from beginning of chromosome to the crossover point is copied from one parent, the rest is copied from the second parent.



- Example :

Parent 1 : X X | X X X X X

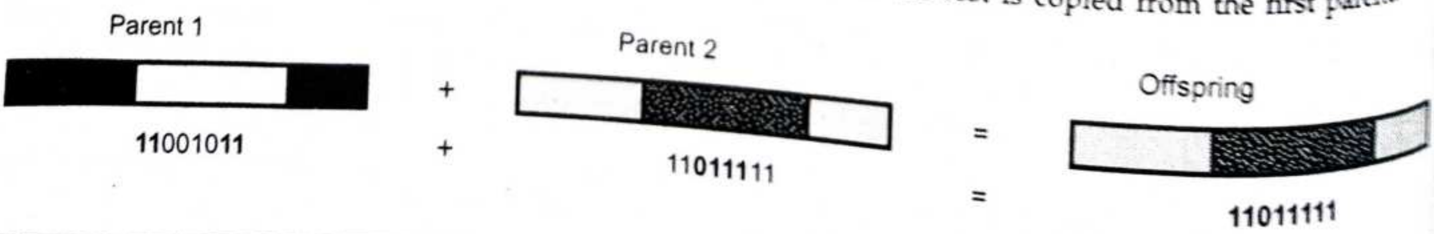
Parent 2 : Y Y | Y Y Y Y Y

Offspring 1 : X X Y Y Y Y Y

Offspring 2 : Y Y X X X X X

Two Point Crossover

- Two-point crossover is very similar to single-point crossover except that two cut-points are randomly generated instead of one.
- Two point crossover : Two crossover points are selected and the part of the chromosome string between these two points is then swapped to generate two children. Binary string from beginning of chromosome to the first crossover point is copied from one parent, the part from the first to the second crossover point is copied from the second parent and the rest is copied from the first parent.



- Example :

  Parent 1 : X X | X X X | X X

  Parent 2 : Y Y | Y Y Y | Y Y

  Offspring 1 : X X Y Y Y X X

  Offspring 2 : Y Y X X X Y Y

Uniform Crossover

- In uniform crossover, a value of the first parent's gene is assigned to the first offspring and the value of the second parent's gene is to the second offspring with a probability value p_c.
- With probability p_c the value of the first parent's gene is assigned to the second offspring and the value of the second parent's gene is assigned to the first offspring.
- Example :

  Parent 1 : X X X X X X X

  Parent 2 : Y Y Y Y Y Y Y

  Offspring 1 : X Y X Y Y X Y

  Offspring 2 : Y X Y X X Y X
- Crossover between 2 good solutions **MAY NOT ALWAYS** yield a better or as good a solution. Since parents are good, probability of the child being good is high. If offspring is not good (poor solution), it will be removed in the next iteration during "Selection".
- One site crossover is more suitable when string length is small while two site crossover is suitable for large strings.
- **Crossover example**

  Parent A 011011

  Parent B 101100
- Mate the parents by splitting each number as shown between the second and third digits (position is randomly selected).

  01*1011 10*1100
- Now combine the first digits of A with the last digits of B and the first digits of B with the last digits of A. This gives you two new offspring.

  011100

  101011
- If these new solutions or offspring, are better solutions than the parent solutions, the system will keep these as more optimal solutions and they will become parents. This is repeated until some condition (for example number of populations or improvement of the best solution) is satisfied.

Matrix Crossover

- Some real problems are naturally suitable for two-dimensional representation. If this kind of problems is to be solved by genetic algorithms, then each possible solution can very conveniently and naturally be conceptually represented as a two-dimensional table.
- Two-dimensional substring crossover generates two offspring chromosomes by choosing only one of the two crossover strategies (horizontal or vertical).

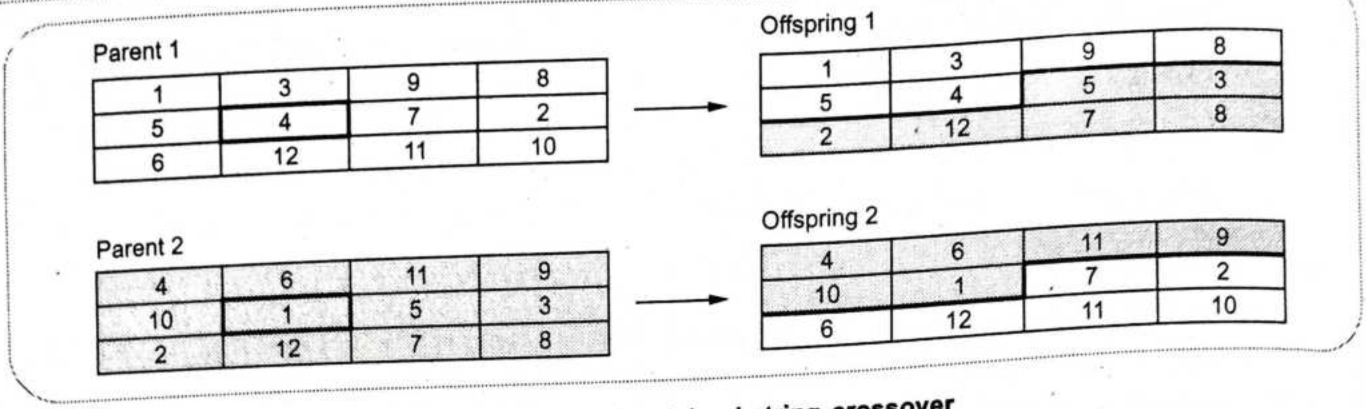


Fig. Q.21.1 Horizontal substring crossover

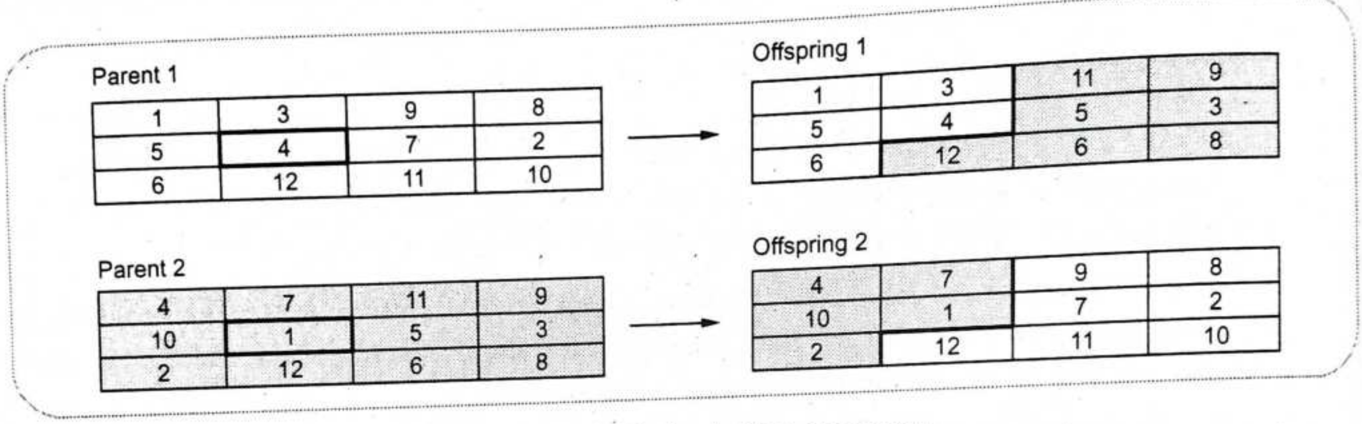


Fig. Q.21.2 Vertical substring crossover

- Alternatively, the two-dimensional crossover operator can be easily modified to generate four offspring chromosomes from a pair of parents by executing the horizontal and the vertical crossovers at the same time.
- The new offspring chromosomes that result from executing the crossover operation may become infeasible for some application problems.

Q.22 What is reproduction ? Give various methods of selecting chromosomes for parents to crossover.

Ans. :

Ans. : • Reproduction is usually the first operator applied on population. Chromosomes are selected from the population to be parents to cross over and produce offspring.

- Reproduction selects good strings in a population and forms a mating pool. This is one of the reasons for the reproduction operation to be sometimes known as the **selection operator.**
- Selection is the process of choosing two parents from the population for crossing. Thus, in reproduction operation the process of natural selection causes those individuals that encode successful structures to produce copies more frequently.
- To sustain the generation of a new population, the reproduction of the individuals in the current population is necessary. For better individuals, these should be from the fittest individuals of the previous population.
- Through **reproduction**, genetic algorithms produce new generations of improved solutions by selecting parents with higher fitness ratings or by giving such parents a greater probability of being contributors and by using random selection.
- If the selection pressure is too low, the convergence rate will be slow and the GA will take unnecessarily longer to find the optimal solution. If the selection pressure is too high, there is an increased change of the GA prematurely converging to an incorrect solution.
- Selection scheme are of two types : **proportionate based selection and ordinal based selection.**

Proportionate Selection

- In proportionate selection, individuals are assigned a probability of being selected based on their fitness :

$$p_i = f_i \sum f_i$$

where p_i is the probability that individual i will be selected,

fi is the fitness of individual i and

$\sum$ fj represents the sum of all the fitnesses of the individuals with the population.

- This type of selection is similar to using a roulette wheel where the fitness of an individual is represented as proportionate slice of wheel. The wheel is then spun and the slice underneath the wheel when it stops determines which individual becomes a parent.
- There are a number of disadvantages associated with using proportionate selection :
  1. Cannot be used on minimization problems,
  2. Loss of selection pressure (search direction) as population converges,
  3. Susceptible to super individuals

Ordinal based selection

In ordinal based selection, individuals are assigned subjective fitness based on the rank within the population :

$$\mathbf{sf_i = (P - r_i)(max - min)/(P - 1) + min}$$

where r_i is the rank of indvidual i,

P is the population size,

Max represents the fitness to assign to the best individual,

Min represents the fitness to assign to the worst individual.

- $p_i = sf_i / \sum sf_i$ Roulette Wheel Selection can be performed using the subjective fitness.
- One disadvantage associated with linear rank selection is that the population must be sorted on each cycle.
- Selection has to balance with variation from crossover and mutation.

1. Too strong selection means sub optimal highly fit individuals will take over the population, reducing the diversity needed for change and progress.
2. Too weak selection will result in too slow evolution.

- The various methods of selecting chromosomes for parents to crossover are :
  1. Roulette-Wheel selection
  2. Boltzmann selection
  3. Tournament selection
  4. Rank selection
  5. Steady state selection

Q.23 Discuss the categorization of bit-wise operator.

Ans. : 1. One's complement operator : Bitwise one's compliment operator will invert the binary bits. If a bit is 1, it will change it to 0. If the bit is 0, it will change it to 1.

2. Logical bitwise operator : there are three logical bitwise operators. They are Bit-wise AND, Bit-wise Exclusive-OR, Bit-wise OR.

- Bitwise AND (&) operator will take two equal length binary sequence and perform the bitwise AND operation on each pair of bit sequence. AND operator will return 1, if both bits are 1. Otherwise it will return 0.

X	0	0	1	1
Y	0	1	0	1
Result	0	0	0	1

- Bitwise XOR (^) operator will take two equal length binary sequence and perform bitwise XOR operation on each pair of bit sequence. XOR operator will return 1, if both bits are different. If bits are same, it will return 0.
- Bitwise OR (|) operator will take two equal length binary sequence and perform bitwise OR operation on each pair of bit sequence. Bitwise OR will return 1, if any of the bit in the pair is 1. If both bits are 0, then the outcome will be 0.

X	0	0	1	1
Y	0	1	0	1
Result	0	1	1	1

3. Shift Operator : Shift operator are of two types, shift left (<<) and shift right (>>).

- Bitwise Left shift (<<) operator is used to shift the binary sequence to the left side by specified position.
- Bitwise Right shift operator (>>) is used to shift the binary sequence to right side by specified position.

4.3 : Genetic Algorithm Cycle

Q.24 Explain the genetic algorithm cycle with example.

Ans. : • GA represents an iterative process. Each iteration is called a **generation.** A typical number of generations for a simple GA can range from 50 to over 500. The entire set of generations is called a **run.**

- Because GAs use a stochastic search method, the fitness of a population may remain stable for a number of generations before a superior chromosome appears.
- A common practice is to terminate a GA after a specified number of generations and then examine the best chromosomes in the population. If no satisfactory solution is found, the GA is restarted.

Convergence Analysis

- The convergence of a genetic algorithm arises when the genes of some high rated individuals quickly attain to dominate the population, constraining it to converge to a local optimum. In this case, the genetic operators cannot produce any more descendents better that the parents; the algorithm ability to continue the search for better solutions is therefore substantially reduced.
- The advantages of genetic algorithms first become apparent when a population of strings is observed.
- Let f be the function $x \rightarrow x^2$ which is to be maximized, in the interval [0, 1]. A population of N numbers in the interval [0, 1] is generated in 10-bit fixed-point coding.
- The function f is evaluated for each of the numbers $x_1, x_2, \ldots, x_N$, and the strings are then listed in descending order of their function values. Two strings from this list are always selected to generate a new member of a new population, whereby the probability of selection decreases monotonically in accordance with the ascending position in the sorted list.
- The computed list contains N strings which, for a sufficiently large N, should look like this :

1 0.1 *************

1 0.1 *************

...

...

1 0.0 *************

- The first positions in the list are occupied by strings in which the first bit after the point is a 1. The last positions are occupied by strings in which the first bit after the decimal point is a 0. The asterisk stands for any bit value from 0 to 1 and the zero in front of the point does not need to be coded in the strings.
- The upper strings are more likely to be selected for a recombination, so that the offspring is more likely to contain a 1 in the first bit than a 0. The new population is evaluated and a new fitness list is drawn up.
- Also the strings with a 0 in the first bit are placed at the end of the list and are less likely to be selected than the strings which begin with a 1. After several generations no more strings with a 0 in the first bit after the decimal point are contained in the population.
- The same process is repeated for the strings with a zero in the second bit. They are also pushed towards extinction. Gradually the whole population converges to the optimal string 0.1111111111.
- With this quadratic function the search operation is carried out within a very well-ordered framework. New points are defined at each crossover, but steps in the direction x = 1 are more likely than steps in the opposite direction. The step length is also reduced in each reproduction step in which the 0 bits are eliminated from a position.

- John Holland suggested the notion of schemata for the convergence analysis of genetic algorithms. Schemata are bit patterns which function as representatives of a set of binary strings. We already used such bit patterns in the example above : the bit patterns can contain each of the three symbols 0, 1 or *. The schema **00**, for example, is a representative of all strings of length 6 with two zeros in the central positions, such as : 100000, 110011, 010010, etc.
- Some authors have argued in favor of the building block hypothesis to explain why GAs do well in some circumstances. According to this hypothesis a GA finds building blocks which are then combined through the generations in order to reach the optimal solution. But the correlations between the optimization parameters sometimes preclude altogether the formation of these building blocks.
- Fig. Q.24.1 shows genetic algorithm cycle. Building blocks are combined together due to combined action of generic operators to form bigger and better building blocks and finally converge to the optimal solution.

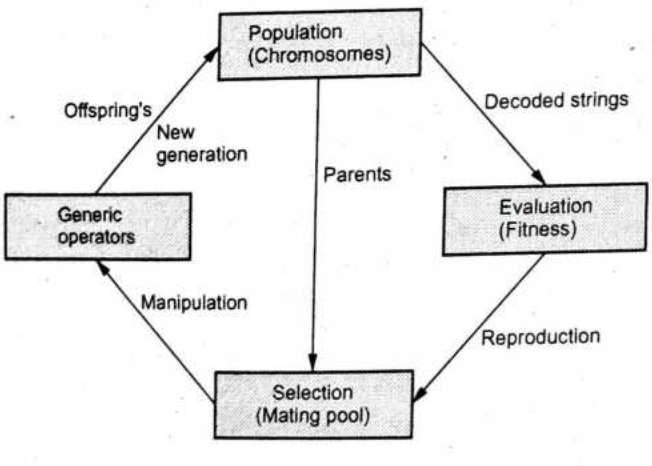


Fig. Q.24.1 Genetic algorithm cycle

4.4 : Fitness Function

Q.25 What is fitness function ? Explain.

Ans. : • Fitness is an important concept in genetic algorithms. The fitness of a chromosome determines how likely it is that it will reproduce. Fitness is usually measured in terms of how well the chromosome solves some goal problem. Fitness can also be subjective (aesthetic). E.g., if the genetic algorithm is to be used to sort numbers, then the fitness of a chromosome will be determined by how close to a correct sorting it produces.

- A fitness function quantifies the optimality of a solution (chromosome) so that particular solution may be ranked against all the other solutions. A fitness value is assigned to each solution depending on how close it actually is to solving the problem.
- Ideal fitness function correlates closely to goal plus quickly computable.
- Example : In TSP, f(x) is sum of distances between the cities in solution. The lesser the value, the fitter the solution is.
- The performance of the individual strings is measured by a fitness function. A fitness function is a problem specific user defined heuristic. After each iteration, the members are given a performance measure derived from the fitness function and the "fittest" members of the population will propagate the next iteration.

$$\text{Fitness} = F_i,\ \text{Hit} = \lambda_i,\ \text{Survival} = \phi_i,$$

$$\text{Death} = \delta_i$$

$$F_i = 2\lambda_i - \delta_i + \sum \phi_i$$

- The fitness function is defined over the genetic representation and measures the ***quality*** of the represented solution. The fitness function is always problem dependent.
- For instance, in the knapsack problem we want to maximize the total value of objects that we can put in a knapsack of some fixed capacity. A representation of a solution might be an array of bits, where each bit represents a different object and the value of the bit (0 or 1) represents whether or not the object is in the knapsack.
- Not every such representation is valid, as the size of objects may exceed the capacity of the knapsack. The *fitness* of the solution is the sum of values of all objects in the knapsack if the representation is valid or 0 otherwise. In some problems, it is hard or even impossible to define the fitness expression; in these cases, interactive genetic algorithms are used.

Q.26 What is role of fitness function in GA ?

Ans. : A fitness function is a type of objective functions which summarizes the goodness of a solution with a single figure of merit.

A fitness function quantifies the optimality of a solution (chromosome) so that particular solution may be ranked against all the other solutions. A fitness value is assigned to each solution depending on how close it actually is to solving the problem.

Q.27 Define the term encoding.

Ans. : Encoding is the process of representing the solution in the form of a string that conveys the necessary information. It is a process of representing individual genes. The process can be performed using bits, numbers, trees, array or any other objects.

Q.28 Explain use of tree encoding.

Ans. : In tree encoding every chromosome is a tree of some objects, such as functions or commands in programming language. It is used in genetic programming.

Q.29 What is encoding ? When to select encoding method ? Explain type of binary encoding methods.

Ans. : • Encoding is the process of representing the solution in the form of a string that conveys the necessary information. It is a process of representing individual genes. The process can be performed using bits, numbers, trees, array or any other objects.

- Just as in a chromosome, each gene controls a particular characteristic of the individual; similarly, each bit in the string represents a characteristic of the solution.
- When choosing an encoding method rely on the following key ideas
  1. Use a data structure as close as possible to the natural representation.
  2. Write appropriate genetic operators as needed.
  3. If possible, ensure that all genotypes correspond to feasible solutions.
  4. If possible, ensure that genetic operators preserve feasibility

Binary Encoding

- Most common method of encoding. Chromosomes are strings of 1s and 0s and each position in the chromosome represents a particular characteristic of the problem. The length of the string is usually determined according to the desired solution accuracy.

Chromosome A 110100011010

Chromosome B 011111111100

- **Octal encoding :** This encoding uses string made up of octal numbers 0 to 7.

Chromosome A 03467216

Chromosome B 15723314

- **Hexadecimal encoding :** This encoding uses string made up of hexadecimal numbers 0 to 9 and A to F.

Chromosome A 9CE7

Chromosome B 3DBA

Permutation Encoding

- Useful in ordering problems such as the Traveling Salesman Problem (TSP). Example : In TSP, every chromosome is a string of numbers, each of which represents a city to be visited.
- Permutation encoding is used in scheduling and ordering problems. Every chromosome is a string of numbers, which represents numbers in a sequence. The kind of encoding usually needs to make some corrections after the crossover and mutation operating procedures because any transformation might create an illegal situation.

Chromosome A 1 5 3 2 6 4 7 9 8

Chromosome B 8 5 6 7 2 3 1 4 9

- **Example of problem :** Traveling Salesman Problem (TSP).
- **The problem :** There are cities and given distances between them. Traveling salesman has to visit all of them, but he does not to travel very much. Find a sequence of cities to minimize traveled distance. **Encoding :** Chromosome says order of cities, in which salesman will visit them.

Value Encoding

- Used in problems where complicated values, such as real numbers, are used and where binary encoding would not suffice. It is good for some problems, but often necessary to develop some specific crossover and mutation techniques for these chromosomes.
- **Example of problem** : Finding weights for neural network.
- **The problem** : There is some neural network with given architecture. Find weights for inputs of neurons to train the network for wanted output.
- **Encoding** : Real values in chromosomes represent corresponding weights for inputs.

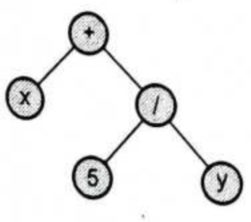
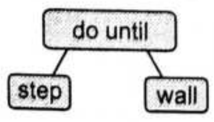
Chromosome A 1.2324 5.3243 0.4556 2.3293 2.4545

Chromosome B ABDJEIFJDHDIERJFDLDFLFEGT

Chromosome C (back), (back), (right), (forward), (left)

Tree Encoding

- In **tree encoding** every chromosome is a tree of some objects, such as functions or commands in programming language. It is used in genetic programming.
- **Example of problem** : Finding a function from given values.
- **The problem** : Some input and output values are given. Task is to find a function, which will give the best (closest to wanted) output to all inputs.
- **Encoding** : Chromosome is functions represented in a tree.

Chromosome A	Chromosome B
Chromosome A  (+x(/5y))	**Chromosome B**  **(do_until step wall)**

4.5 : Applications of Genetic Algorithm

Q.30 Mention the application area of genetic algorithms.

Ans. : • Genetic algorithms are global optimization techniques that utilize concurrent search from multiple-points rather than from a single-point. It is independent of the problem complexity. The main necessity of the GA is to specify the objective function and to place finite bounds on the parameters. GA is widely used for robust power system stabilization.

- Optimization using GA techniques are widely applied in many real world problems such as image processing, pattern recognition, classifiers, machine learning.

Traveling Salespersons Problem

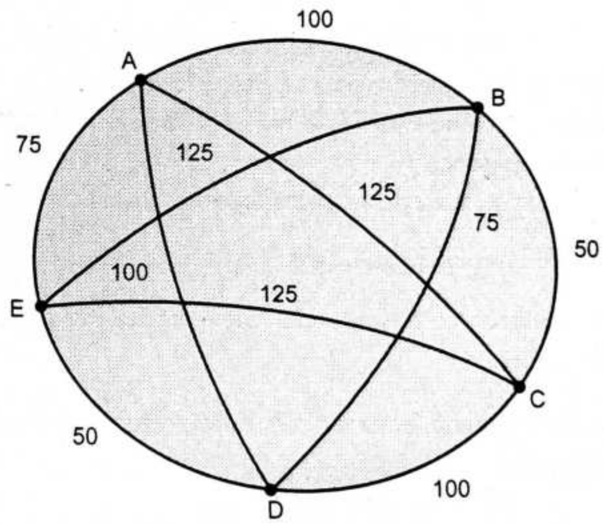


Fig. Q.30.1

- Suppose a salesperson has five cities to visit and then must return home.
- The goal of the problem is to find the shortest path for the salesperson to travel.
- The traveling salesman must visit every city in his territory exactly once (possibly then return to the starting point). Given the cost of travel between all cities, how should he plan his itinerary for minimum total cost of the entire tour ?

Representation :

- Representation is an ordered list of city numbers known as an *order-based* GA.

1) Mumbai 3) Surat 5) Pune 7) Nagpur

2) Calcutta 4) Delhi 6) Bangalore 8) Chennai

CityList1	(3 5 7 2 1 6 4 8)
CityList2	(2 5 7 6 8 1 3 4)

- Crossover combines inversion and recombination :

Parent1	(3 5 7 2 1 6 4 8)
Parent2	**(2 5 7 6 8 1 3 4)**
Child	(5 8 7 2 1 6 3 4)

This operator is called the *Order1* crossover.

- Mutation involves reordering of the list :

	* *
Before :	(5 8 7 2 1 6 3 4)
After :	(5 8 6 2 1 7 3 4)

Traveling Salesperson problem with 9 city

- Now, the problem is how to crossover ?

P1 = (192465783)

P2 = (459187623)

- First of all, select two cut point, indicate by a "|", which are randomly inserted into the same location of each parent.

P1 = (192 | 4657 | 83)

P2 = (459 | 1876 | 23)

Two children C1 and C2 are produced in the following way.

First, the segments between cut points are copied into the offspring :

C1 = (XXX | 4657 | XX)

C2 = (XXX | 1876 | XX)

Next, starting from the second cut point of one parent, the cities from the other parent are copied in the same order, omitting cities already present. When the end of the string is reached, continue from the beginning.

Thus, the sequence of cities from P2 (459 | 1876 | 23) is 23 459 1876.

For C1= (XXX | 4657 | XX), once 4657 are removed from the sequence generated by P2, we get the sequence 23918.

Then we just use these numbers to fill in the XXX XX portion in order.

Thus, C1 = (239 | 4657 | 18)

Q.31 Discuss the applications of genetic algorithm : match word finding.

Ans. : Problem is to 'guess' a word.

- Difficult to solve with a Brute force approach. Assuming case sensitivity, to investigate every possible combination of letters in a seven letter word would require 1,028,071,702,528 attempts. Assume trying to guess the word 'genetic'. Also assign fitness based on number of correct letters in the correct place.
- **Step one :** Select an encoding schema (use characters)
- **Step two :** Initialize the population

1. kdjirid 2. ginddcc 3. nmugjyb
4. zezezez 5. uhjklyt 6. wojikli
7. kladonn 8. flortik

Step three : Evaluate the population

Individual	Genotype	Fitness
1	kdjirid	1
2	ginddcc	3
3	nmugjyb	0
4	zezezez	2
5	uhjklyt	0
6	wojikli	0
7	kladonn	0
8	flortik	2

Average fitness = 1

Step four : Select breeding population and selection based on fitness, so breeding population is,

Individual	Genotype	Fitness
2	ginddcc	3
2	ginddcc	3
2	ginddcc	3
4	zezezez	2
4	zezezez	2
8	flortik	2
8	flortik	2
1	kdjirid	1

Step five : Create new population

Here crossover comes into play and no mutation used here to keep things simple. For example : cross individual 2 with individual 4.

individual 2 genotype is : ginddcc

individual 4 genotype is : zezezez

crossing over produces : genedec

- New population is :

Individual	Genotype	Fitness
1	genedec	5
2	glnrdic	4
3	fdoitik	2
4	zdziziz	1
5	gdnidic	4
6	feoetek	3
7	kijdrcd	0
8	zlzrez	0

- **Average fitness = 2.375**

Fill in the Blanks for Mid Term Exam

Q.1 The chromosome are divided into several parts called ______.

Q.2 Genetic algorithms are inspired by ______ theory about evolution.

Q.3 The mutation depends on the encoding as well as the ______.

Q.4 A genetic algorithm is a _____ technique used in computing to find true or approximate solutions to optimization and search problems.

Q.5 ______ is the process of taking two parent solutions and producing from them a child.

Multiple Choice Questions for Mid Term Exam

Q.1 Genetic Algorithm are a part of ______.

a evolutionary computing

b inspired by Darwin's theory about evolution - "survival of the fittest"

c are adaptive heuristic search algorithm based on the evolutionary ideas of natural selection and genetics

d all of the above

Q.2 Which GA operation is computationally most expensive ?

a Initial population creation.

b Selection of sub-population for mating.

c Reproduction to produce next generation.

d Convergence testing.

Q.3 Which of the following is not true for Genetic algorithms ?

a It is a probabilistic search algorithm.

b It is guaranteed to give global optimum solutions.

c If an optimization problem has more than one solution, then it will return all the solutions.

d It is an iterative process suitable for parallel programming.

Q.4 The purpose of the fitness evaluation operation is ________.

a to check whether all individual satisfies the constraints given in the problem.

b to decide the termination point.

c to select the best individuals.

d to identify the individual with worst cost function.

Q.5 Roulette wheel selection scheme is preferable when ________.

a fitness values are uniformly distributed.

b fitness values are non-uniformly distributed.

c needs low selection pressure.

d needs high population diversity

Q.6 Tournament selection has ________.

a low population diversity and moderate selection pressure

b high population diversity and Moderate selection pressure

c moderate population diversity and high selection pressure

d high population diversity and low selection pressure

Q.7 The purpose of the fitness evaluation operation is ________.

a to check whether all individual satisfies the constraints given in the problem.

b to decide the termination point.

c to select the best individuals.

d to identify the individual with worst cost function.

Q.8 Choose the example for certain-outcome problem.

a 8-puzzle

b Bridge

c Tic-tac-toc

d Chess

Answer Keys for Fill in the Blanks :

Q.1	genes	Q.2	darwin's
Q.3	crossover	Q.4	search
Q.5	Crossover		

Answer Keys for Multiple Choice Questions :

Q.1	d	Q.2	c
Q.3	b	Q.4	c
Q.5	a	Q.6	c
Q.7	c	Q.8	a

END...

UNIT - V

5 Rough Sets

5.1 : Introduction of Rough Sets

Q.1 What is set and rough set ?

Ans. : Set :

- A set of objects that possesses similar characteristics it is a fundamental part of mathematics. All the mathematical objects, such as relations, functions and numbers can be considered as a set.
- The various components of a set are known as elements, and relationship between an element and a set is called of a pertinence relation. Cardinality is the way of measuring the number of elements of a set.

Rough set :

- In the rough set theory, membership is not the primary concept. Rough sets represent a different mathematical approach to vagueness and uncertainty.
- The rough set methodology is based on the premise that lowering the degree of precision in the data makes the data pattern more visible.

Q.2 Define reducts.

Ans. : The process of reducing an information system such that the set of attributes of the reduced information system is independent and no attribute can be eliminated further without losing some information from the system, the result is known as reduct.

Q.3 What is rough approximation ? What is lower and upper approximation ? List the properties of approximation.

Ans. :

- Rough set concept can be defined quite generally by means of topological operations, interior and closure, called **approximations.**
- Suppose we are given a set of objects U called the universe and an indiscernibility relation $R \subseteq U \times U$, representing our lack of knowledge about elements of U. For the sake of simplicity we assume that R is an equivalence relation.
- Let X is a subset of U. We want to characterize the set X with respect to R. To this end we will need the basic concepts of rough set theory given below.
  1. The lower approximation of a set X with respect to R is the set of all objects, which can be for certain classified as X with respect to R (are certainly X with respect to R).
  2. The upper approximation of a set X with respect to R is the set of all objects which can be possibly classified as X with respect to R (are possibly X in view of R).
  3. The boundary region of a set X with respect to R is the set of all objects, which can be classified neither as X nor as not-X with respect to R.
- We are ready to give the definition of rough sets.
  1. Set X is crisp (exact with respect to R), if the boundary region of X is empty
  2. Set X is rough (inexact with respect to R), if the boundary region of X is nonempty.
- *A set is said to be rough if its boundary region is non-empty, otherwise the set is crisp.*
- Formal definitions of approximations and the boundary region are as follows :
- R-lower approximation of X

$$R_*(x) = \bigcup_{x \in U} \{R(x) : R(x) \subseteq X\}$$

- R-upper approximation of X

$$R^*(x) = \bigcup_{x \in U} \{R(x) : R(x) \cap X \neq \phi\}$$

- R-boundary region of X

$$RN_R(X) = R^*(x) - R_*(X)$$

- The **lower approximation** of a set is union of all granules which are entirely included in the set; the **upper approximation** is union of all granules which have non-empty intersection with the set; the **boundary region** of set is the difference between the upper and the lower approximation.
- Fig. Q.3.1 shows approximations.

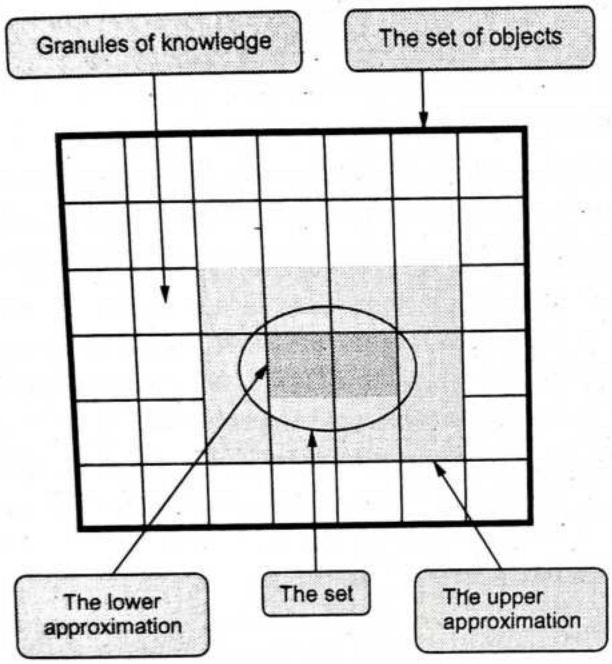


Fig. Q.3.1 Approximations

- Approximations have the following properties :

1) $R_*(X) \subseteq X \subseteq R^*(X)$
2) $R_*(\phi) = R^*(\phi) = \phi.\ R_*(U) = R^*(U) = U$
3) $R^*(X \cup Y) = R^*(X) \cup R^*(Y)$
4) $R_*(X \cap Y) = R_*(X) \cap R_*(Y)$
5) $R_*(X \cup Y) \supseteq R_*(X) \cup R_*(Y)$
6) $R^*(X \cap Y) \subseteq R^*(X) \cap R^*(Y)$
7) $X \subseteq Y \rightarrow R_*(X) \subseteq R_*(Y)\ \&\ R^*(X) \subseteq R^*(Y)$
8) $R_*(-X) = -R^*(X)$
9) $R^*(-X) = R_*(X)$
10) $R_*R_*(X) = R^*R_*(X) = R_*(X)$
11) $R^*R^*(X) = R_*R^*(X) = R^*(X)$

Q.4 Explain indiscernibility relation with example.

Ans. :

- Indiscernibility relation is a central concept in rough set theory, and is considered as a relation between two objects or more, where all the values are identical in relation to a subset of considered attributes.
- A decision system (i.e. a decision table) expresses the entire model. This table may be unnecessarily large. The same or indiscernible objects may be represented several times.
- An information system or information table can be viewed as a table, consisting of objects (rows) and attributes (columns). It is used in the representation of data that will be utilized by rough set, where each object has a given amount of attributes.
- These objects are described in accordance with the format of the data table, in which rows are considered objects for analysis and columns as attributes. Below is shown an example of information.

Patient	Attributes			
	Headache	Vomiting	Temperature	Viral illness
#1	No	Yes	High	Yes
#2	Yes	No	High	Yes
#3	Yes	Yes	Very high	Yes
#4	No	Yes	Normal	No
#5	Yes	No	High	No
#6	No	Yes	Very high	Yes

Table Q.4.1

- It can be observed that the set is composed of attributes that are directly related to the patients' symptoms whether they be headache, vomiting and temperature.
- When Table Q.4.1 is broken down it can be seen that the set regarding {patient2, patient3, patient5} is indiscernible in terms of headache attribute.
- The set concerning {patient1, patient3, patient4} is indiscernible in terms of vomiting attribute. Patient2 has a viral illness, whereas patient5 does not,

however they are indiscernible with respect to the attributes headache, vomiting and temperature. Therefore, patient2 and patient5 are the elements of patients' set with unconcluded symptoms.

- A **binary relation** $R \subseteq X \times X$ which is **reflexive** (i.e. an object is in relation with itself xRx), **symmetric** (if xRy then yRx) and **transitive** (if xRy and yRz then xRz) is called an **equivalence relation.**
- The **equivalence class** of an element $x \in X$ consists of all objects $y \in X$ such that **xRy.**
- Let A = (U, A) be an information system then with any $B \subseteq A$ there is associated an ***equivalence relation*** IND_A ***(B) :***

$$IND_A(B) = \{(x, x') \in U2 \mid a \in B \; a(x) = a(x')\}$$

- $IND_A(B)$ is called the ***B-indiscernibility relation.*** If $(x, x') \in IND_A(B)$ then x and x' are indiscernible from each other by attributes from B. The equivalence classes of the ***B-indiscernibility*** relation are denoted $[x]_B$.

Q.5 Consider the following table

U	*Headache*	*Temp.*	*Flu*
U1	Yes	*Normal*	*No*
U2	Yes	*High*	*Yes*
U3	Yes	*Very-high*	*Yes*
U4	No	*Normal*	*No*
U5	No	*High*	*No*
U6	No	*Very-high*	*Yes*
U7	No	*High*	*Yes*
U8	No	*Very-high*	*No*

Find indiscernibility classes.

Ans. : The indiscernibility classes defined by R = {Headache, Temp.}

= {u1}, {u2}, {u3}, {u4}, {u5, u7}, {u6, u8}.

X1 = {u | Flu(u) = yes}

= {u2, u3, u6, u7}

Lower approximation $\underline{R}X1$ = {u2, u3}

Upper approximation $\overline{R}X1$ = {u2, u3, u6, u7, **u8, u5**}

Q.6 Explain advantages of rough set approach

Ans. : • It does not need any preliminary or additional information about data ? like probability in statistics, grade of membership in the fuzzy set theory.

- It provides efficient methods, algorithms and tools for finding hidden patterns in data.
- It allows to reduce original data, i.e. to find minimal sets of data with the same knowledge as in the original data.
- It allows to evaluate the significance of data.
- It allows to generate in automatic way the sets of decision rules from data.
- It is easy to understand.
- It offers straightforward interpretation of obtained results.
- It is suited for concurrent (parallel/distributed) processing.

5.2 : Rule Induction and Discernibility Matrix

Q.7 What is discernability matrix ?

Ans. : • Two objects are discernible if their values are different in at least one attribute. Discernability matrix is a basic method to search all the reducts of an information system.

- A discernability matrix is a square matrix in which rows and columns are objects, and cells are attribute sets that discern objects.
- Given an information table S, its discernability matrix M = (M(x,y)) is a $|U| \times |U|$ matrix, in which the element M(x,y) for an object pair (x,y) is defined by :

$$M(x,y) = \{a \in At \mid I_a(x) \neq I_a(y)\}$$

- The physical meaning of the matrix element M(x,y) is that objects x and y can be distinguished by any attribute in M(x,y). The pair (x,y) can be discerned if $M(x,y) \neq \varnothing$.
- A discernability matrix M is symmetric, i.e., M(x,y) = M(y,x) and $M(x,x) = \varnothing$. Therefore, it is sufficient to consider only the lower triangle or the upper triangle of the matrix.

Q.8 Define minimum discernability matrix.

Ans. : • Given a reduct R, for all M(x,y) ≠ ∅, we have R ∩ M(x,y) ≠ ∅. Thus, for each non - empty M(x,y), we can replace it by one and only one attribute in R to preserve the original discernability of x and y. Such a simplest form is called a minimum discernability matrix. For a given reduct R, we may obtain more than one minimum discernability matrix.

- A discernability matrix can be transformed to a minimum matrix by the application of a finite number of elementary matrix operations.

Q.9 What is reducts ?

Ans. : • An information system built on real world data, always contains redundant information, such as repeated rows, indiscernible rows, dependable attributes. To infer a more concise equivalent information system is very useful in practice.

- The equivalent information system has fewer rows and columns and can be considered a pattern of the original one. In rough set theory, we call this pattern a reduct.
- The process to obtain reducts is based on two concepts : 1) indiscernibility relation, and 2) set of approximation.
- Indiscernibility relation resolves the row redundancy, whereas approximation handles column redundancy

Q.10 What is rule induction ?

Ans. : • Rule induction is one of the most important techniques of machine learning. Since regularities hidden in data are frequently expressed in terms of rules, rule induction is one of the fundamental tools of data mining at the same time.

- Usually, rules are expressions of the form

if (attribute – 1, value – 1) and (attribute – 2, value – 2) and ··· and (attribute – n, value – n) then (decision, value).

- Some rule induction systems induce more complex rules, in which values of attributes may be expressed by negation of some values or by a value subset of the attribute domain.
- Data from which rules are induced are usually presented in a form similar to a table in which cases (or examples) are labels (or names) for rows and variables are labeled as attributes and a decision.

Q.11 Define global and local rule induction.

Ans. : In global rule induction algorithms, the search space is the set of all attribute values, while in local rule induction algorithms the search space is the set of attribute-value pairs.

5.3 : Integration of Soft Computing Techniques

Q.12 Explain difference between neural processing and fuzzy processing.

Ans. :

Neural processing	Fuzzy processing
It tries to incorporate human thinking process to solve problems without mathematically modeling.	Fuzzy logic allows making definite decisions based on imprecise or ambiguous data.
Learning is start from scratch.	Priori knowledge is required.
Learning algorithm is used.	Learning algorithm is not used.
Behavior is like black box.	Simple to implement.
Neural networks are good at recognizing patterns, they are not good at explaining how they reach their decisions.	Fuzzy logic can reason with imprecise information, are good at explaining their decisions but they cannot automatically acquire the rules they use to make those decisions.

Q.13 What is Neuro-fuzzy system ?

Ans. : A neuro-fuzzy system is a neural network which is functionally equivalent to a fuzzy inference model. It can be trained to develop IF-THEN fuzzy rules and determine membership functions for input and output variables of the system.

Q.14 What is use of fuzzification layer ?

Ans. : Neurons in this layer represent fuzzy sets used in the antecedents of fuzzy rules. A fuzzification neuron receives a crisp input determines the degree to which this input belongs to the neuron's fuzzy set.

Q.15 Can genetic algorithms help us in selecting the network architecture ?

Ans. : The architecture of the network often determines the success or failure of the application. Usually the network architecture is decided by trial and error; there is a great need for a method of automatically designing the architecture for a particular application. Genetic algorithms may well be suited for this task.

Q.16 What are the characteristics of Neuro-fuzzy hybrid system ?

Ans. :

1. A neuro-fuzzy system based on an underlying fuzzy system is trained by means of a data-driven learning method derived from neural network theory.
2. It can be represented as a set of fuzzy rules at any time of the learning process, i.e., before, during and after.
3. System may be initialized with or without prior knowledge in terms of fuzzy rules.
4. The learning procedure is constrained to ensure the semantic properties of the underlying fuzzy system.
5. A neuro-fuzzy system approximates n-dimensional unknown function which is partly represented by training examples.

Q.17 How does a neuro-fuzzy system learn ?

Ans. : • A neuro-fuzzy system is essentially a multi-layer neural network and thus it can apply standard learning algorithms developed for neural networks, including the back-propagation algorithm.

- When a training input-output example is presented to the system, the back-propagation algorithm computes the system output and compares it with the desired output of the training example.
- The error is propagated backwards through the network from the output layer to the input layer. The neuron activation functions are modified as the error is propagated.
- To determine the necessary modifications, the back-propagation algorithm differentiates the activation functions of the neurons.
- A neuro-fuzzy system is a fuzzy system that uses a learning algorithm derived from or inspired by neural network theory to determine its parameters by processing data samples.

Q.18 Write short note on genetic neuro hybrid system.

Ans. : • Genetic algorithm is an adaptive search technique used for solving mathematical problems and engineering optimization problems. Genetic algorithm attempts to find a good solution to the problem by genetically breeding a population of individuals over a series of generations.

- Artificial neural networks and genetic algorithms are both abstractions of natural processes. They are formulated into a computational model so that the learning power of neural networks and adaptive capabilities of evolutionary processes can be combined.
- Back-propagation is one method to train the network. The training is performed by one pattern at a time. The training of all patterns of a training set is called an epoch. The training set has to be representative collections of input-out-put examples.
- Back-propagation training is a gradient descent algorithm. It tries to improve the performance of the neural net by reducing its error along its gradient. The error (E) is the half the sum of the geometric averages of the difference between projected target (t) and the actual output (o) vector over all patterns (p). In each training step, the weights (w) are adjusted towards the direction of maximum decrease, scaled by some learning rate lambda.
- The basic GA operators are crossover, selection and mutation. Information about the neural network is encoded in the genome of the genetic algorithm. Fig. Q.10.1 shows structure of genetic neuro hybrid system.
- The hybrid systems optimize neural networks in two ways : evolution by the genetic algorithm and learning by back-propagation.
- The weights that are improved by the evaluation process are coded back into the chromosome. This undermines the ability of the genetic algorithm to perform hyper-plane sampling, since the codes are

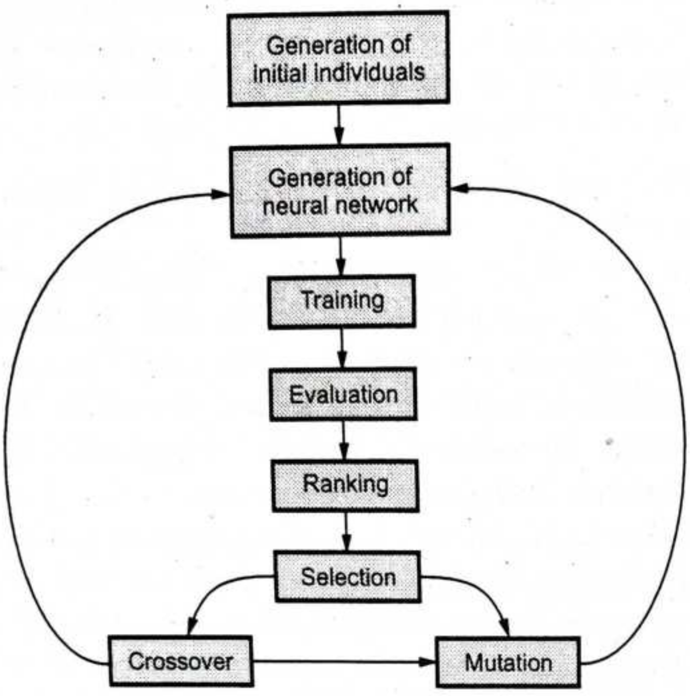


Fig. Q.18.1

altered during evaluation. Still, in the case of cellular encoding, it improves the hybrid system.

- The improved weights are discarded after evaluation; the learning guides the evolutionary search. If an encoded network is close to an optimum, back-propagation will reach this optimum, thus resulting in a good fitness value. Since the bit-strings that encode networks that are close to one optimum share a lot of bit patterns, the genetic algorithm is able to exploit these by hyper plane sampling.
- It requires large amount of memory to handle and manipulates chromosomes.

Q.19 What is back propagation network ?

Ans. : Back propagation network is a method of training multi-layer neural network. Backpropagation is a supervised learning algorithm, for training Multi-layer Perceptrons.

Q.20 Define intuitionistic fuzzy rough sets.

Ans. : If A = (A_L, A_U) and B = (B_L, B_U) are two fuzzy rough sets in X with $B \subseteq \overline{A}$, then the ordered pair (A, B) is called an intuitionistic fuzzy rough set in X. The condition $B \subseteq \overline{A}$ is called the intuitionistic condition.

Fill in the Blanks for Mid Term Exam

Q.1 Discernibility matrix is a basic method to search all the ________ of an information system.

Q.2 ________ is the way of measuring the number of elements of a set.

Q.3 In global rule induction algorithms, the search space is the set of all attribute values, while in local rule induction algorithms the search space is the set of ________ pairs

Q.4 A fuzzy system-based NFS is trained by means of ——— learning method derived from neural network theory.

Q.5 BPN is a method of teaching ——— neural networks how to perform a given task.

Q.6 An output membership neuron combines all its inputs by using the fuzzy operation union. This operation can be implemented by the ——— .

Q.7 ——— hybrid system uses pipeline concept.

Q.8 Neuro-fuzzy system is an example of ——— hybrid system.

Multiple Choice Questions for Mid Term Exam

Q.1 Discernibility matrix is a ______ matrix in which rows and columns are objects, and cells are attribute sets that discern objects.

a zero

b square

c non-zero

d binary

Q.2 Which of the following is not a hybrid system ?

a Embedded hybrid system

b Sequential hybrid system

c Auxiliary hybrid system

d Parallel hybrid system

Q.3 Fuzzy - Genetic Hybrid system is a _______.

a Fuzzy logic in parallel with the Genetic algorithm

b Fuzzy logic controlled Genetic algorithm

c Genetic algorithm controlled Fuzzy logic

d None of the above

Q.4 What are the applications of Fuzzy Inference Systems ?

a Wireless services, heat control and printers

b Restrict power usage, telephone lines and sort data

c Simulink, boiler and CD recording

d Automatic control, decision analysis and data classification

Q.5 What are the following sequence of steps taken in designing a fuzzy logic machine ?

a Fuzzification->Rule evaluation->Defuzzification

b Rule evaluation->Fuzzification->Defuzzification

c Fuzzy Sets->Defuzzification->Rule evaluation

d Defuzzification->Rule evaluation->Fuzzification

Answer Keys for Fill in the Blanks :

Q.1	reducts	Q.2	Cardinality
Q.3	attribute-value	Q.4	data-driven
Q.5	multi-layer	Q.6	probabilistic OR
Q.7	Sequential	Q.8	embedded

Answer Keys for Multiple Choice Questions :

Q.1	b	Q.2	d
Q.3	b	Q.4	d
Q.5	a		

END...

SOLVED MODEL QUESTION PAPER

Soft Computing [As per R18 Pattern]

B.Tech., IV - I [CSE / IT] Professional Elective - V

Time : 3 Hours] **[Max. Marks : 75**

Answer any five questions
All questions carry equal marks

Q.1 a) *List and explain in brief constituents of soft computing.* **(Refer Q.11 of Chapter - 1)**

b) *What type of problem is solved by using soft computing ?* **(Refer Q.15 of Chapter - 1)** [7 + 8]

Q.2 a) *Give the properties of fuzzy sets and also explain the operations involved in it.* **(Refer Q.3 of Chapter - 2)**

b) *What is meant by fuzzy logic system ? Illustrate it with proper examples.* **(Refer Q.20 of Chapter - 2)** [7 + 8]

Q.3 a) *Explain flow diagram showing the particle swarm optimization algorithm with Pseudo code.* **(Refer Q.11 of Chapter - 3)**

b) *Explain global best (gbest) and local best (lbest) particle swarm optimization.* **(Refer Q.13 of Chapter - 3)** [8 + 7]

Q.4 a) *Explain the genetic algorithm cycle with example.* **(Refer Q.24 of Chapter - 4)**

b) *How does genetic algorithm differ from conventional algorithm ? Give the advantages of GA over conventional algorithms.* **(Refer Q.12 of Chapter - 4)** [9 + 6]

Q.5 a) *Write short note on genetic neuro hybrid system.* **(Refer Q.18 of Chapter - 5)**

b) *Explain indiscernibility relation with example.* **(Refer Q.4 of Chapter - 5)** [8 + 7]

Q.6 *What is defuzzification ? Explain different methods of defuzzification.* **(Refer Q.34 of Chapter - 2)** [15]

Q.7 *What is encoding ? When to select encoding method ? Explain type of binary encoding methods.* **(Refer Q.29 of Chapter - 4)** [15]

Q.8 a) *List and explain methods employed for membership value assignment.* **(Refer Q.14 of Chapter - 2)**

b) *What is rough approximation ? What is lower and upper approximation ? List the properties of approximation.* **(Refer Q.3 of Chapter - 5)** [6 + 9]

END...